



DATA BOOK FOR HEAT TRANSFER APPLICATIONS

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NOMENCLATURE

c_p	: Specific heat, heat capacity (kJ/kg K)
Gr	: Grashof number
k	: Thermal conductivity (W/m K)
M	: Molar mass (kg/kmol)
Nu	: Nusselts number
P	: Pressure (Pa)
Pr	: Prandtl number
P_{sat}	: Saturation pressure
R	: Gas constant (kJ/kg K)
Re	: Reynolds number
Ra	: Rayleighs number
T	: Temperature (K)
v	: Specific volume, molar volume ($m^3/kmol$)
α	: Thermal diffusivity (m^2/s)
β	: Coefficient of thermal expansion ($1/K$)
δ	: Boundary layer thickness (m)
ϵ	: Emissivity
μ	: Dynamic viscosity (kg/m s, Pa s)
ν	: Kinematic viscosity (m^2/s)
ρ	: Density (kg/m^3)
h_{fg}	: Enthalpy of vaporization (kJ/kg)
ϵ	: Heat exchanger effectiveness
St	: Stanton number
Bi	: Biot number
Fo	: Fourier number
τ	: Time (s)
h	: Convective heat transfer coefficient ($W/m^2 K$)

Subscripts for Temperatures

b	: bulk temperature
f	: film temperature
w	: wall temperature
∞	: free stream temperature

SECTION A**THERMO PHYSICAL PROPERTIES****CONTENTS**

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TABLE A.1
MOLAR MASS, GAS CONSTANT AND CRITICAL PROPERTIES OF
SELECTED SUBSTANCES

Substance	<i>M</i> kg/kmol	<i>R</i> kJ/kg K	Properties at Critical point		
			<i>T</i> K	<i>P</i> MPa	<i>v</i> m ³ /kmol
Air	28.970	0.28700	132.5	3.770	0.0883
Ammonia (NH ₃)	17.030	0.48820	405.5	11.280	0.0724
Argon	39.948	0.20810	151.0	4.860	0.0749
Benzene (C ₆ H ₆)	78.115	0.10640	562.0	4.920	0.2603
Bromine	159.808	0.05200	584.0	10.340	0.1355
n – Butane (C ₄ H ₁₀)	58.124	0.14300	425.2	3.800	0.2547
Carbon dioxide	44.010	0.18890	304.2	7.390	0.0943
Carbon monoxide	28.011	0.29680	133.0	3.500	0.0930
Carbon tetrachloride (CCl ₄)	153.820	0.05405	556.4	4.560	0.2759
Chlorine	70.906	0.11730	417.0	7.710	0.1242
Chloroform (CHCl ₃)	119.380	0.06964	536.6	5.470	0.2403
Dichlorodifluoromethane R-12 (CCl ₂ F ₂)	120.910	0.06876	384.7	4.010	0.2179
Dichlorofluoromethane R-12 (CHCl ₂ F)	102.920	0.08078	451.7	5.170	0.1973
Ethane (C ₂ H ₆)	30.070	0.27650	305.5	4.480	0.1480
Ethyl alcohol (C ₂ H ₅ OH)	46.070	0.18050	516.0	6.380	0.1673
Ethylene (C ₂ H ₄)	28.054	0.29640	282.4	5.120	0.1242
Helium	4.003	2.07690	5.3	0.230	0.0578
n – Hexane (C ₆ H ₁₄)	86.179	0.09647	507.9	3.030	0.3677
Hydrogen (normal)	2.016	4.12400	33.3	1.300	0.0649
Krypton	83.800	0.09921	209.4	5.500	0.0924
Methane (CH ₄)	16.043	0.51820	191.1	4.640	0.0993
Methyl alcohol (CH ₃ OH)	32.042	0.25950	513.2	7.950	0.1180
Methyl Chloride (CH ₃ Cl)	50.488	0.16470	416.3	6.680	0.1430
Neon	20.183	0.41190	44.5	2.730	0.0417
Nitrogen	28.013	0.29680	126.2	3.390	0.0899
Nitrous oxide (N ₂ O)	44.013	0.18890	309.7	7.270	0.0961
Oxygen	31.999	0.25980	154.8	5.080	0.0780
Propane (C ₃ H ₈)	44.097	0.18850	370.0	4.260	0.1998
Propylene (C ₃ H ₆)	42.081	0.19760	365.0	4.620	0.1810
Sulfur dioxide	64.063	0.12980	430.7	7.880	0.1217
Tetrafluoroethane (R-134a) (CF ₃ CH ₂ F)	102.030	0.08149	374.3	4.067	0.1847
Trichlorofluoromethane (R-11) (CCl ₃ F)	137.370	0.06052	471.2	4.380	0.2478
Water	18.015	0.46150	647.3	22.090	0.0568
Xenon	131.300	0.06332	289.8	5.880	0.1186

TABLE A.2
BOILING AND FREEZING POINT PROPERTIES OF SELECTED
SUBSTANCES

Substance	Boiling data at 1 atm		Freezing data		Liquid properties		
	NBP	LHV	FP	LHI	T	ρ	c_p
	K	kJ/kg	K	kJ/kg	K	kg/m ³	kJ/kg K
Ammonia	239.85	1357.0	195.45	322.4	239.85	682.0	4.430
					253.15	665.0	4.520
					273.15	639.0	4.600
					298.15	602.0	4.800
Argon	87.25	161.6	83.85	28.0	210.60	1394.0	1.140
Benzene	353.35	394.0	278.65	126.0	293.15	879.0	1.720
Brine (20% NaCl by mass)	377.05	---	255.75	---	293.15	1150.0	3.110
n-Butane	272.65	385.2	134.65	80.3	272.65	601.0	2.310
Carbon dioxide	194.75	230.5 (at 0°C)	216.55	---	273.15	298.0	0.590
Ethanol	351.35	838.3	158.95	109.0	298.15	783.0	2.460
Ethylene glycol	471.25	800.1	262.35	181.1	293.15	1109.0	2.840
Ethyl alcohol	351.75	855.0	117.15	108.0	293.15	789.0	2.840
Glycerine	453.05	974.0	292.05	200.6	293.15	1261.0	2.320
Helium	4.25	22.8	---	---	4.25	146.2	22.800
Hydrogen	20.35	445.7	13.95	59.5	20.35	70.7	10.000
Isobutane	261.45	367.1	113.15	105.7	261.45	593.8	2.280
Kerosene	477.15-566.15	251.0	248.25	---	293.15	820.0	2.000
Mercury	629.85	294.7	234.25	11.4	298.15	13560.0	0.139
Methane	111.65	510.4	90.95	58.4	111.65	423.0	3.490
					173.15	301.0	5.790
Methanol	337.65	1100.0	175.45	99.2	298.15	787.0	2.550
Nitrogen	77.35	198.6	63.15	25.3	77.35	809.0	2.060
					113.15	596.0	2.970
Octane	397.95	306.3	215.65	180.7	293.15	703.0	2.100
Oil (light)	---	---	---	---	298.15	910.0	1.800
Oxygen	90.15	212.7	54.35	13.7	90.15	1141.0	1.710
Petroleum	---	230-284	---	---	293.15	640.0	2.000
Propane	231.05	427.8	85.45	80.0	231.05	581.0	2.250
					273.15	529.0	2.530
					323.15	449.0	3.130
R 134a	247.05	216.8	176.55	---	223.15	1443.0	1.230
					247.05	1374.0	1.270
					273.15	1294.0	1.340
					298.15	1206.0	1.420
Water	373.15	2257.0	273.15	333.7	273.15	1000.0	4.230
					298.15	997.0	4.180
					323.15	988.0	4.180
					348.15	975.0	4.190
					373.15	958.0	4.220

NBP : Normal boiling point

FP : Freezing point

LHV : Latent heat of vaporization, h_{fg} LHI : Latent heat of fusion, h_{if}

TABLE A.3
PROPERTIES OF SELECTED SOLID METALS

Material	Melting point	Properties at 300 K					Thermal conductivity (k, W/m K) and Specific heat (c _p , J/kg K) at various temperatures								
		K	ρ	c _p	k	α	k	c _p	100 K	200 K	400 K	600 K	800 K	1000 K	
			kg/m ³	J/kg K	W/m K	× 10 ⁶ m ² /s									
Aluminum and its alloys															
	Pure	933	2702	903	237.00	97.10	k	302.0	237.00	240.0	231.0	218.0			
							c _p	482.0	798.00	949.0	1033.0	1146.0			
Alloy 2024 – T6 (4.5% Cu, 1.5% Mg, 0.6% Mn)	775	2770	875	177.00	73.00	k	65.0	163.00	186.0	186.0	---	---			
						c _p	473.0	787.00	925.0	1042.0	---				
Alloy 195, cast (4.5%)	---	2790	883	168.00	68.20	---	---	---	174.0	185.0	---				
Beryllium	1550	1850	1825	200.00	59.20	k	990.0	301.00	161.0	126.0	106.0	90.80			
						c _p	203.0	1114.00	2191.0	2604.0	2823.0	3018.00			
Bismuth	545	9780	122	7.86	6.59	k	16.5	9.69	7.04	---	---	---			
						c _p	112.0	120.00	127.0	---	---				
Boron	2573	2500	1107	27.00	9.76	k	190.0	55.50	16.8	10.6	9.6.0	9.85			
						c _p	128.0	600.00	1463.0	1892.0	2160.0	2338.00			
Cadmium	594	8650	231	96.80	48.40	k	203.0	99.30	94.7	---	---	---			
						c _p	198.0	222.00	242.0	---	---				
Chromium	2118	7160	449	93.70	29.10	k	159.0	111.00	90.9	80.7	71.3	65.40			
						c _p	192.0	384.00	484.0	542.0	581.0	616.00			
Cobalt	1769	8862	421	99.20	26.60	k	167.0	122.00	85.4	67.4	58.2	52.10			
						c _p	236.0	379.00	450.0	503.0	550.0	628.00			
Copper and its alloys															
	Pure	1358	8933	385	401.00	117.00	k	482.0	413.00	393.0	379.0	366.0	352.00		
							c _p	252.0	356.00	397.0	417.0	433.0	451.00		

Continued..

Table A.3 PROPERTIES OF SELECTED SOLID METALS

Material	Melting point	Properties at 300 K					Thermal conductivity (k, W/m K) and Specific heat (c _p , J/kg K) at various temperatures							
		ρ kg/m ³	c _p J/kg K	k W/m K	α × 10 ⁵ m ² /s	k	c _p	100 K	200 K	400 K	600 K	800 K	1000 K	
Commercial bronze (90% Cu, 10% Al)	K	8800	420	52.0	14.00	k	c _p	---	42.0	52.0	59.0	---	---	
Phosphor gear bronze (89% Cu, 11% Sn)	1104	8780	355	54.0	17.00	k	c _p	---	41.0	65.0	74.0	---	---	
Cartridge brass (70% Cu, 30% Zn)	1188	8530	380	110.0	33.90	k	c _p	75.0	95.0	137.0	149.0	---	---	
Constantan (55% Cu, 45% Ni)	1493	8920	384	23.0	6.71	k	c _p	---	360.0	395.0	425.0	---	---	
Germanium	1211	5360	322	59.9	34.70	k	c _p	237.0	362.0	---	---	---	---	
Gold	1336	19300	129	317.0	127.00	k	c _p	190.0	290.0	337.0	348.0	357.0	375.0	
Iridium	2720	22500	130	147.0	50.30	k	c _p	109.0	124.0	131.0	135.0	140.0	145.0	
Iron:	1810	7870	447	80.2	23.10	k	c _p	134.0	94.0	69.5	54.7	43.3	32.8	
Pure	---	7870	447	72.7	20.70	k	c _p	216.0	384.0	490.0	574.0	680.0	975.0	
Armco (99.75% pure)	---	7870	447	72.7	20.70	k	c _p	95.6	80.6	65.7	53.1	42.2	32.3	
Carbon Steels:	---	7854	434	60.5	17.70	k	c _p	215.0	384.0	490.0	574.0	680.0	975.0	
Plain (Mn ≤1%, Si ≤0.1%)	---	7854	434	60.5	17.70	k	c _p	---	---	56.7	48.0	39.2	30.0	
						c _p		---	---	487.0	559.0	685.0	1169.0	

Continued..

Table A.3 PROPERTIES OF SELECTED SOLID METALS

Material		Melting point	Properties at 300 K					Thermal conductivity (k, W/m K) and Specific heat (c_p , J/kg K) at various temperatures									
			ρ kg/m ³	c_p J/kg K	k W/m K	α $\times 10^6$ m ² /s	k	100 K	200 K	400 K	600 K	800 K	1000 K				
														c_p			
Carbon – silicon (Mn $\leq 1\%$, 0.1 % < Si $\leq 0.6\%$)		K	7817	446	51.9	14.90	k	---	---	---	---	---	---	---	---	---	---
Carbon – Manganese – Silicon (1% < Mn $\leq 6.65\%$, 0.1% < Si $\leq 0.6\%$)		---	8131	434	41.0	11.60	k	---	---	---	---	---	---	---	---	---	---
							c_p										
Chromium Steels:																	
0.5 Cr – 0.25 Mo – Si (0.18% C, 0.65% Cr, 0.23% Mo, 0.6% Si)		---	7822	444	37.7	10.90	k	---	---	---	---	---	---	---	---	---	---
							c_p										
1Cr – 0.5Mo (0.16% C, 1% Cr, 0.54% Mo, 0.39% Si)		---	7858	442	42.3	12.20	k	---	---	---	---	---	---	---	---	---	---
							c_p										
1Cr-V (0.2% C, 1% Cr, 0.15% V)		---	7836	443	48.9	14.10	k	---	---	---	---	---	---	---	---	---	---
							c_p										
Stainless Steels:																	
AISI 302		---	8055	480	15.1	3.91	k	---	---	---	---	---	---	---	---	---	---
							c_p										
AISI 304		1670	7900	477	14.9	3.95	k	9.2	12.6	16.6	19.8	22.6	25.4	25.4	25.4	25.4	25.4
							c_p	272.0	402.0	515.0	557.0	582.0	611.0	611.0	611.0	611.0	611.0
AISI 316		---	8238	468	13.4	3.48	k	---	---	---	---	---	---	---	---	---	---
							c_p	---	---	---	---	---	---	---	---	---	---
AISI 347		---	7978	480	14.2	3.71	k	---	---	---	---	---	---	---	---	---	---
							c_p	---	---	---	---	---	---	---	---	---	---
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Table A.3 PROPERTIES OF SELECTED SOLID METALS

Material	Melting point	Properties at 300 K					Thermal conductivity (k , W/m K) and Specific heat (c_p , J/kg K) at various temperatures							
		ρ kg/m ³	c_p J/kg K	k W/m K	α $\times 10^6$ m ² /s	k	c_p	100 K	200 K	400 K	600 K	800 K	1000 K	
Lead	601	11340	129	35.3	24.1		k	39.7	36.7	34.0	31.4	----	----	
Magnesium	923	1740	1024	156.0	87.6		c_p	118.0	125.0	132.0	142.0	----	----	
Molybdenum	2894	10240	251	138.0	53.7		c_p	649.0	934.0	1074.0	1170.0	1267.0	----	
Nickel and its alloys							k	179.0	143.0	134.0	126.0	118.0	112.0	
							c_p	141.0	224.0	261.0	275.0	285.0	295.0	
Pure	1728	8900	444	90.7	23.0		k	164.0	107.0	80.2	65.6	67.6	71.8	
Nichrome (80% Ni, 20% Cr)	1672	8400	420	12.0	3.4		c_p	232.0	383.0	485.0	592.0	530.0	562.0	
							k	----	----	14.0	16.0	21.0	----	
Inconel X-750 (73% Ni, 15% Cr, 6.7% Fe)	1665	8510	439	11.7	3.1		c_p	----	----	480.0	525.0	545.0	----	
							k	8.7	10.3	13.5	17.0	20.5	24.0	
Niobium	2741	8570	265	53.7	23.6		c_p	----	372.0	473.0	510.0	546.0	626.0	
							k	55.2	52.6	55.2	58.2	61.3	64.4	
Palladium	1827	12020	244	71.8	24.5		c_p	188.0	249.0	274.0	283.0	292.0	301.0	
							k	76.5	71.6	73.6	79.7	86.9	94.2	
Platinum and its alloys							c_p	168.0	227.0	251.0	261.0	271.0	281.0	
Pure	2045	21450	133	71.6	25.1		k	77.5	72.6	71.8	73.2	75.6	78.7	
							c_p	100.0	125.0	136.0	141.0	146.0	152.0	
Alloy (60% Pt, 40% Rh)	1800	16630	162	47.0	17.4		k	----	----	52.0	59.0	65.0	69.0	
							c_p	----	----	----	----	----	----	

Continued..

Table A.3 PROPERTIES OF SELECTED SOLID METALS

Material	Melting point	Properties at 300 K					Thermal conductivity (k, W/m K) and Specific heat (c _p , J/kg K) at various temperatures							
		ρ kg/m ³	c _p J/kg K	k W/m K	α ×10 ⁶ m ² /s	k	c _p	100 K	200 K	400 K	600 K	800 K	1000 K	
														K
Rhenium	3453	21100	136	47.9	16.7		c _p	58.9	51.0	46.1	44.2	44.1	44.6	
						k		97.0	127.0	139.0	145.0	151.0	156.0	
Rhodium	2236	12450	243	150.0	49.6		c _p	186.0	154.0	146.0	136.0	127.0	121.0	
						k		147.0	220.0	253.0	274.0	293.0	311.0	
Silicon	1685	2330	712	148.0	89.2		c _p	884.0	264.0	98.9	61.9	42.4	31.2	
						k		259.0	556.0	790.0	867.0	913.0	946.0	
Silver	1235	10500	235	429.0	174.0		c _p	444.0	430.0	425.0	412.0	396.0	379.0	
						k		187.0	225.0	239.0	250.0	262.0	277.0	
Tantalum	3269	16600	140	57.5	24.7		c _p	59.2	57.5	57.8	58.6	59.4	60.2	
						k		110.0	133.0	144.0	146.0	149.0	152.0	
Thorium	2023	11700	118	54.0	39.1		c _p	59.8	54.6	54.5	55.8	56.9	56.9	
						k		99.0	112.0	124.0	134.0	145.0	156.0	
Tin	505	7310	227	66.6	40.1		c _p	85.2	73.3	62.2	----	----	----	
						k		188.0	215.0	243.0	----	----	----	
Titanium	1953	4500	522	21.9	9.32		c _p	30.5	24.5	20.4	19.4	19.7	20.7	
						k		300.0	465.0	551.0	591.0	633.0	675.0	
Tungsten	3660	19300	132	174.0	68.3		c _p	208.0	186.0	159.0	137.0	125.0	118.0	
						k		87.0	122.0	137.0	142.0	146.0	148.0	
Uranium	1406	19070	116	27.6	12.5		c _p	21.7	25.1	29.6	34.0	38.8	43.9	
						k		94.0	108.0	125.0	146.0	176.0	180.0	
Vanadium	2192	6100	489	30.7	10.3		c _p	35.8	31.3	31.3	33.3	35.7	38.2	
						k		258	430.0	515.0	540.0	563.0	597.0	
Zinc	693	7140	389	116.0	41.8		c _p	117.0	118.0	111.0	103.0	---	---	
						k		297.0	367.0	402.0	436.0	---	---	
Zirconium	2125	6570	278	22.7	12.4		c _p	33.2	25.2	21.6	20.7	21.6	23.7	
						k		205.0	264.0	300.0	332.0	342.0	362.0	

TABLE A.4
PROPERTIES OF SELECTED NONMETAL SOLIDS

Material	Melting point	Properties at 300K					Thermal conductivity (k, W/m K) and Specific heat (c _p , J/kg K) at various temperatures						
		ρ kg/m ³	c _p J/kg K	k W/m K	α × 10 ⁶ m ² /s	k	c _p	100 K	200 K	400 K	600 K	800 K	1000 K
Aluminum oxide, sapphire	K	2323	3970	765	46.00	15.100	k	450.00	82.00	32.4	18.9	13.00	10.50
Aluminum oxide, polycrystalline	K	2323	3970	765	36.00	11.900	k	133.00	55.00	26.4	15.8	10.40	7.85
Beryllium oxide	K	2725	3000	1030	272.00	88.000	k	---	---	940.0	1110.0	1180.00	1225.00
Boron	K	2573	2500	1105	27.60	9.990	k	190.00	52.50	18.7	11.3	8.10	6.30
Pyroceram, Corning 9606	K	1623	2600	808	3.98	1.890	k	5.25	4.78	3.64	3.28	3.08	2.96
Silicon Carbide	K	3100	3160	675	490.00	230.000	k	---	---	908.0	1038.0	1122.00	1197.00
Silicon dioxide, polycrystalline (fused silica)	K	1883	2220	745	1.38	0.834	k	0.69	1.14	1.51	1.75	2.17	2.87
Silicon Nitride	K	2173	2400	691	16.00	9.650	k	---	---	905.0	1040.0	1105.00	1155.00
Sulfur	K	392	2070	708	0.206	0.141	k	0.165	0.185	---	---	---	---
Thorium dioxide	K	3573	9110	235	13.00	6.100	k	403.00	606.00	---	---	---	---
Titanium dioxide, polycrystalline	K	2133	4157	710	8.40	2.800	k	---	---	7.01	5.02	3.94	3.46

Continued..

Table A.4 PROPERTIES OF SELECTED NONMETAL SOLIDS

Material	Melting point K	Properties at 300K					Thermal conductivity (k, W/m K) and Specific heat (c_p , J/kg K) at various temperatures					
		ρ	c_p	k	α	k	100 K	200 K	400 K	600 K	800 K	1000 K
		kg/m ³	J/kg K	W/m K	$\times 10^6$ m ² /s		c_p					
Titanium dioxide, polycrystalline	2133	4157	710	8.40	2.8	k	k	---	7.01	5.02	3.94	3.46
						c_p	c_p	---	805.00	880.00	910.00	930.00
Boron fiber epoxy (30% volume composite)												
parallel to fibers	590	2080		2.29		k	k	2.10	2.23	2.28	---	---
perpendicular to fibers	590	2080		0.59		k	k	0.37	0.49	0.60	---	---
			1122			c_p	c_p	364.00	757	1431.00	---	---
Carbon:												
Amorphous	1500	1950	---	1.60	---	k	k	0.67	1.18	1.89	2.37	2.53
Diamond, type IIa insulator	---	3500	509	2300.00	---	k	k	10000.00	4000.00	1540.00	---	---
						c_p	c_p	21.00	194.00	853.00	---	---
Graphite, pyrolytic	2273	2210										
parallel to layers				1950.00		k	k	4970.00	3230.00	1390.00	667.00	534.00
perpendicular to layers				5.70		k	k	16.80	9.23	4.09	2.68	2.01
			709			c_p	c_p	136.00	411.00	992.00	1406.00	1793.00
Graphite fiber epoxy (25% volume)	450	1400										
parallel to fibers				11.10		k	k	5.70	8.70	13.00	---	---
perpendicular to fibers				0.87		k	k	0.46	0.68	1.10	---	---
			935			c_p	c_p	337.00	642.00	1216.00	---	---
Silicon Dioxide Crystalline (Quartz)												
parallel to fibers	1883	2650		10.40		k	k	39	16.4	7.6	5.00	4.20
perpendicular to fibers	1883	2650		6.21		k	k	20.8	9.5	4.7	3.40	3.10
			745			c_p	c_p	---	---	885	1075.00	1250.00

TABLE A.5
PROPERTIES OF INSULATING, BUILDING AND OTHER MATERIALS

Substance	T	k	ρ	c_p	α
	K	W/m K	kg/m ³	kJ/kg K	$\times 10^7 \text{ m}^2/\text{s}$
Asphalt	293.15	0.700	2120	0.92	3.60
Brick:					
carborundum	1673.15	11.100	---	---	---
common	293.15	0.38 - 0.52	1800	0.84	2.8 - 3.4
chrome	373.15	1.900	---	---	---
facing	293.15	1.300	---	---	---
fire clay brick	773.15	1.040	2000	0.96	5.40
magnesite (50% MgO)	293.15	2.680	2000	---	---
masonry	293.15	0.660	1700	0.84	4.60
silica (95% SiO ₂)	293.15	1.070	1900	---	---
zircon (62% ZrO ₂)	293.15	2.440	3600	---	---
Cement (Portland, fresh, dry)	293.15	0.300	3100	0.75	1.30
Cement mortar	296.15	1.160	---	---	---
Coal, anthracite	303.15	0.260	1300	1.25	1.60
Concrete, cinder	296.15	0.760	---	---	---
Concrete made with gravel, dry	293.15	1.280	2200	0.88	6.60
Glass window	293.15	0.780	2700	0.84	3.40
Glass borosilicate	303.75 - 348.15	1.090	2200	---	---
Particle board:					
low density	303.15	0.079	590	1.30	1.00
high density	303.15	0.170	1000	1.30	1.30
Plaster:					
gypsum	293.15	0.480	1440	0.84	4.00
metal lath	293.15	0.470	---	---	---
wood lath	293.15	0.280	---	---	---
Polyethylene	303.15	0.330	960	2.10	1.64
Polypropylene	303.15	0.160	1150	1.90	0.73
Polyvinylchloride	303.15	0.090	1700	1.10	0.48

Continued..

Table A.5 PROPERTIES OF INSULATING, BUILDING AND OTHER MATERIALS

Substance	T K	k W/m K	ρ kg/m ³	c_p kJ/kg K	α $\times 10^7$ m ² /s
Rubber, hard	303.15	0.150	1200	2.000	0.62
Silicon carbide	303.15	490.00	3150	0.680	2290.00
Stone:					
granite	---	1.73 - 3.98	2640	0.820	8 - 18
lime stone	373.15 - 673.15	1.26 - 1.33	2500	0.900	5.6 - 5.9
marble	---	2.07 - 2.94	2500 - 2700	0.800	10 - 13.6
sand stone	313.15	1.830	2160 - 2300	0.710	11.2 - 11.9
Teflon	303.15	0.350	2200	1.050	1.5
Wood (across the grain):					
balsa	303.15	0.055	140	---	---
cypress	303.15	0.097	460	---	---
fir	296.15	0.110	420	2.720	0.96
maple or oak	303.15	0.166	540	2.400	1.28
yellow pine	296.15	0.147	640	2.800	0.82
white pine	303.15	0.112	430	---	---
Acoustic tile	303.15	0.060	290	1.300	1.60
Asbestos:					
loosely packed	228.15	0.149	---	---	---
	273.15	0.154	470 - 570	0.816	3.3 - 4
	373.15	0.161	---	---	---
asbestos - cement boards	293.15	0.740	---	---	---
asbestos - cement sheets	324.15	0.166	---	---	---
felt, 16 laminations per cm	311.15	0.057	---	---	---
	423.15	0.069	---	---	---
	533.15	0.083	---	---	---
felt, 8 laminations per cm	311.15	0.078	---	---	---
	423.15	0.095	---	---	---
	533.15	0.112	---	---	---

Continued..

Table A.5 PROPERTIES OF INSULATING, BUILDING AND OTHER MATERIALS

Substance	T K	k W/m K	ρ kg/m ³	c_p kJ/kg K	α $\times 10^7$ m ² /s
Corrugated	311.15	0.087	---	---	---
	366.15	0.100	---	---	---
	423.15	0.119	---	---	---
Balsam wood	305.15	0.040	35	---	---
Card board, corrugated	---	0.064	---	---	---
Celotex	305.15	0.048	---	---	---
Cork board	303.15	0.043	160	---	---
Cork, regranulated	305.15	0.045	45 - 120	1.88	2 - 5.3
Cork ground	305.15	0.043	150	---	---
Diatomaceous earth	273.15	0.061	320	---	---
Felt, hair	303.15	0.036	130 - 200	---	---
Felt, wool	303.15	0.052	330	---	---
Fiber, insulating board	293.15	0.048	240	---	---
Glass wool	296.15	0.038	24	0.70	22.6
Glass fiber, duct liner	303.15	0.038	32	0.84	14.1
Glass fiber, loose blown	303.15	0.043	16	0.84	32.0
Ice	273.15	2.220	910	1.93	12.6
Insulux, dry	305.15	0.064	---	---	---

Continued..

Table A.5 PROPERTIES OF INSULATING, BUILDING AND OTHER MATERIALS

Substance	T	k	ρ	c_p	α
	K	W/m K	kg/m ³	kJ/kg K	$\times 10^7$ m ² /s
Kapok	303.15	0.035	---	---	---
Magnesia, 85%	311.15	0.067	270	---	---
	366.15	0.071	---	---	---
	423.15	0.074	---	---	---
	477.15	0.080	---	---	---
Rock wool	305.15	0.040	160	---	---
Rock wool loosely packed	423.15	0.067	64	---	---
	533.15	0.087	---	---	---
Saw dust	296.15	0.059	---	---	---
Silica aerogel	305.15	0.024	140	---	---
Styrofoam	305.15	0.033	---	---	---
Wood shavings	296.15	0.059	---	---	---

TABLE A.6
PROPERTIES OF SATURATED WATER

T K	P _{sat} kPa	ρ kg/m ³		h _{fg} kJ/kg	c _p J/kg K		k W/m K		μ kg/m s		Pr		β 1/K
		Liquid	Vapour		Liquid	Vapour	Liquid	Vapour	Liquid $\times 10^3$	Vapour $\times 10^5$	Liquid	Vapour	
273.15	0.6113	999.8	0.0048	2501	4217	1854	0.561	0.0171	1.792	0.922	13.5	1	-0.068
278.15	0.8721	999.9	0.0068	2490	4205	1857	0.571	0.0173	1.519	0.934	11.2	1	0.015
283.15	1.2276	999.7	0.0094	2478	4194	1862	0.58	0.0176	1.307	0.946	9.45	1	0.733
288.15	1.7051	999.1	0.0128	2466	4186	1863	0.589	0.0179	1.138	0.959	8.09	1	0.138
293.15	2.3390	998.0	0.0173	2454	4182	1867	0.598	0.0182	1.002	0.973	7.01	1	0.195
298.15	3.1690	997.0	0.0231	2442	4180	1870	0.607	0.0186	0.891	0.987	6.14	1	0.247
303.15	4.2460	996.0	0.0304	2431	4178	1875	0.615	0.0189	0.798	1.001	5.42	1	0.294
308.15	5.6280	994.0	0.0397	2419	4178	1880	0.623	0.0192	0.720	1.016	4.83	1	0.337
313.15	7.3840	992.1	0.0512	2407	4179	1885	0.631	0.0196	0.653	1.031	4.32	1	0.377
318.15	9.5930	990.1	0.0655	2395	4180	1892	0.637	0.0200	0.596	1.046	3.91	1	0.415
323.15	12.350	988.1	0.0831	2383	4181	1900	0.644	0.0204	0.547	1.062	3.55	1	0.451
328.15	15.7600	985.2	0.1045	2371	4183	1908	0.649	0.0208	0.504	1.077	3.25	1	0.484
333.15	19.9400	983.3	0.1304	2359	4185	1916	0.654	0.0212	0.467	1.093	2.99	1	0.517
338.15	25.0300	980.4	0.1614	2346	4187	1926	0.659	0.0216	0.433	1.110	2.75	1	0.548
343.15	31.1900	977.5	0.1983	2334	4190	1936	0.663	0.0221	0.404	1.126	2.55	1	0.578
348.15	38.5800	974.7	0.2421	2321	4193	1948	0.667	0.0225	0.378	1.142	2.38	1	0.607
353.15	47.3900	971.8	0.2935	2309	4197	1962	0.670	0.0230	0.355	1.159	2.22	1	0.653
358.15	57.8300	968.1	0.3536	2296	4201	1977	0.673	0.0235	0.333	1.176	2.08	1	0.670
363.15	70.1400	965.3	0.4235	2283	4206	1993	0.675	0.0240	0.315	1.193	1.96	1	0.702
368.15	84.5500	961.5	0.5045	2270	4212	2010	0.677	0.0246	0.297	1.210	1.85	1	0.716
373.15	101.3300	957.9	0.5978	2257	4217	2029	0.679	0.0251	0.282	1.227	1.75	1	0.750
383.15	143.2700	950.6	0.8263	2230	4229	2071	0.682	0.0262	0.255	1.261	1.58	1	0.798
393.15	198.5300	943.4	1.1210	2203	4244	2120	0.683	0.0275	0.232	1.296	1.44	1	0.858

Continued..

2007-2022
5-11-2022
5-11-2022

Table A.6 PROPERTIES OF SATURATED WATER

T K	P _{sat} kPa	ρ kg/m ³		h _{fg} kJ/kg	c _p J/kg K		k kW/m K		μ kg/m s		Pr		β 1/K
		Liquid	Vapour		Liquid	Vapour	Liquid	Vapour	Liquid $\times 10^3$	Vapour $\times 10^5$	Liquid	Vapour	
403.15	270.1	934.60	1.496	2174	4263	2177	0.684	0.0288	0.213	1.330	1.330	1.01	0.913
413.15	361.3	921.70	1.965	2145	4286	2244	0.683	0.0301	0.197	1.365	1.240	1.02	0.970
423.15	475.8	916.60	2.546	2114	4311	2314	0.682	0.0316	0.183	1.399	1.160	1.02	1.025
433.15	617.8	907.40	3.256	2083	4340	2420	0.680	0.0331	0.170	1.434	1.090	1.05	1.145
443.15	791.7	897.70	4.119	2050	4370	2490	0.677	0.0347	0.160	1.468	1.030	1.05	1.178
453.15	1002.1	887.30	5.153	2015	4410	2590	0.673	0.0364	0.150	1.502	0.983	1.07	1.210
463.15	1254.4	876.40	6.388	1979	4460	2710	0.669	0.0382	0.142	1.537	0.947	1.09	1.280
473.15	1553.8	864.30	7.852	1941	4500	2840	0.663	0.0401	0.134	1.571	0.910	1.11	1.350
493.15	2318.0	84.03	11.600	1859	4610	3110	0.650	0.0442	0.122	1.641	0.865	1.15	1.520
513.15	3344.0	813.70	16.730	1767	4760	3520	0.632	0.0487	0.111	1.712	0.836	1.24	1.720
533.15	4688.0	783.70	23.690	1663	4970	4070	0.609	0.0540	0.102	1.788	0.832	1.35	2.000
553.15	6412.0	750.80	33.150	1544	5280	4835	0.581	0.0605	0.094	1.870	0.854	1.49	2.380
573.15	8581.0	713.80	46.150	1405	5750	5980	0.548	0.0695	0.086	1.965	0.902	1.69	2.950
593.15	11274.0	667.10	64.570	1239	6540	7900	0.508	0.0836	0.078	2.084	1.000	1.97	
613.15	14586.0	610.50	92.620	1028	8240	11870	0.469	0.1100	0.070	2.255	1.230	2.43	
633.15	18651.0	528.30	144.000	720	14690	25800	0.427	0.1780	0.050	2.571	2.060	3.73	
647.29	22090.0	317.00	317.000	---	∞	∞	∞	∞	0.043	4.313			

TABLE A.7
PROPERTIES OF SELECTED LIQUID METALS

Name of metal; [Melting point (K), Boiling point (K)]	T	ρ	ν	Pr	k	α	c_p
	K	kg/m ³	$\times 10^6$ m ² /s		W/m K	$\times 10^3$ m ² /s	kJ/kg K
Bismuth [544.15, 1763.15]	573.15	10030	0.171	0.0198	13.03	0.008611	0.1507
	673.15	9910	0.142	0.0146	14.42	0.009722	0.1507
	773.15	9785	0.122	0.0113	15.82	0.010833	0.1507
	873.15	9660	0.108	0.0091	17.21	0.011944	0.1507
Lithium [461.15, 1590.15]	473.15	515	1.110	0.0643	37.22	0.017222	4.1868
	573.15	505	0.927	0.0503	38.96	0.018333	4.1868
	673.15	495	0.817	0.0404	41.87	0.020278	4.1868
	773.15	484	0.734	0.0328	45.36	0.0225	4.1868
Mercury [312.05, 630.15]	293.15	13550	0.114	0.0272	7.91	0.004361	0.1390
	373.15	13350	0.094	0.0192	8.96	0.004889	0.1373
	423.15	13230	0.086	0.0162	9.65	0.005306	0.1373
	473.15	13120	0.080	0.0140	10.35	0.005722	0.1373
	573.15	12880	0.071	0.0107	11.75	0.006639	0.1373
Potassium [337.15, 760.15]	422.15	807.3	0.461	0.0066	45.00	0.0699	0.8000
	420.15	741.7	0.240	0.0034	39.50	0.0707	0.7500
	977.15	674.4	0.191	0.0029	33.30	0.0655	0.7500
Sodium [97.3, 878]	423.15	916.0	0.594	0.0087	84.90	0.068333	1.3565
	473.15	903.0	0.506	0.0075	81.41	0.067778	1.3272
	573.15	878.0	0.394	0.0063	70.94	0.063056	1.2812
	673.15	854.0	0.330	0.0056	63.97	0.058889	1.2728
	773.15	829.0	0.289	0.0053	56.99	0.054167	1.2728
Tin [505.05, 2543.15]	523.15	6980.0	0.270	0.0141	34.08	0.019167	0.2554
	573.15	6940.0	0.240	0.0126	33.73	0.019028	0.2554
	673.15	6865.0	0.200	0.0106	33.15	0.018889	0.2554
	773.15	6790.0	0.173	0.0092	32.56	0.01875	0.2554
Alloy (56.5% Bi + 43.5% Pb) [396.65, 1873.15]	423.15	10550.0	0.289	0.0450	9.77	0.006389	0.1465
	473.15	10490.0	0.243	0.0364	10.35	0.006667	0.1465
	573.15	10360.0	0.187	0.0250	11.40	0.007500	0.1465
	673.15	10240.0	0.157	0.0187	12.56	0.010000	0.1465
	773.15	10120.0	0.136	0.0144	13.96	0.009444	0.1465
Alloy (25% Na+75% K) [262.15, 1057.15]	373.15	847.0	0.607	0.0248	23.84	0.024444	1.1430
	473.15	822.0	0.452	0.0171	23.26	0.026389	1.0718
	573.15	799.0	0.366	0.0134	22.68	0.027500	1.0383
	673.15	775.0	0.308	0.0108	22.10	0.028611	1.0060
	773.15	751.0	0.267	0.0090	21.52	0.029722	0.9672
Alloy (45% Na+55% K) [292.15]	366.15	887.4	0.652	0.0260	25.6	0.025556	1.1300
	644.15	821.7	0.287	0.0091	27.5	0.031667	1.0550
	977.15	740.1	0.217	0.0058	28.9	0.037500	1.0430

TABLE A.8
PROPERTIES OF SELECTED SATURATED LIQUIDS

T	ρ	c_p	k	α	ν	p_r
K	kg/m ³	kJ/kg K	$\times 10^3$ W/m K	$\times 10^6$ m ² /s	$\times 10^6$ m ² /s	
AMMONIA						
223.15	704	4.463	546.6	0.174	0.435	2.60
233.15	691	4.467	546.6	0.178	0.406	2.28
243.15	679	4.476	548.9	0.180	0.387	2.15
253.15	667	4.509	546.6	0.182	0.381	2.09
263.15	653	4.564	543.1	0.183	0.378	2.07
273.15	640	4.635	539.6	0.182	0.373	2.05
283.15	626	4.714	531.5	0.180	0.368	2.04
293.15	612	4.798	521.0	0.178	0.358	2.02
303.15	596	4.890	507.1	0.174	0.350	2.01
313.15	581	4.999	493.1	0.170	0.340	2.00
323.15	561	5.116	475.7	0.166	0.330	1.99
CARBON DIOXIDE						
223.15	1156	1.842	85.5	0.040	0.119	2.96
233.15	1120	1.884	101.1	0.048	0.118	2.46
243.15	1077	1.968	111.7	0.053	0.117	2.22
253.15	1032	2.052	115.0	0.054	0.115	2.12
263.15	983	2.177	109.9	0.051	0.113	2.20
273.15	927	2.470	104.6	0.046	0.108	2.38
283.15	863	3.140	97.1	0.036	0.102	2.80
293.15	772	5.024	87.2	0.022	0.091	4.10
303.15	598	36.425	70.3	0.003	0.079	28.70
GLYCERIN						
273.15	1276	2.261	282.6	0.098	8314.000	84700
283.15	1270	2.320	283.8	0.096	3000.000	31000
293.15	1264	2.387	286.1	0.095	1180.000	12500
303.15	1258	2.445	286.1	0.093	501.000	5380
313.15	1252	2.512	286.1	0.091	223.000	2450
323.15	1245	2.583	287.3	0.089	149.000	1630
SULPHUR DIOXIDE						
223.15	1561	1.361	241.9	0.114	0.484	4.24
233.15	1537	1.361	234.9	0.113	0.423	3.74
243.15	1513	1.361	230.3	0.112	0.370	3.31
253.15	1488	1.361	225.6	0.111	0.324	2.93
263.15	1463	1.361	218.6	0.110	0.288	2.62
273.15	1438	1.365	211.7	0.108	0.257	2.38
283.15	1412	1.365	204.7	0.107	0.232	2.18
293.15	1386	1.365	198.9	0.105	0.209	2.00
303.15	1359	1.365	191.9	0.104	0.189	1.83
313.15	1329	1.369	184.9	0.102	0.173	1.70

Continued..

Table A.8 PROPERTIES OF SELECTED SATURATED LIQUIDS

T	ρ	c_p	k	α	ν	p_r
K	kg/m ³	kJ/kg K	$\times 10^3$ W/m K	$\times 10^6$ m ² /s	$\times 10^6$ m ² /s	
323.15	1299	1.369	176.8	0.100	0.162	1.61
METHYL CHLORIDE						
223.15	1052	1.474	215.2	0.139	0.319	2.31
233.15	1033	1.482	209.3	0.137	0.317	2.32
243.15	1016	1.491	202.4	0.134	0.314	2.35
253.15	999	1.503	195.4	0.130	0.309	2.38
263.15	981	1.520	187.2	0.126	0.305	2.43
273.15	962	1.537	177.9	0.121	0.302	2.49
283.15	942	1.562	171.0	0.117	0.293	2.55
293.15	923	1.587	162.8	0.111	0.292	2.63
303.15	903	1.616	153.5	0.106	0.288	2.72
313.15	883	1.650	144.2	0.100	0.281	2.83
323.15	861	1.687	133.8	0.093	0.274	2.97
REFRIGERANT - 12						
223.15	1547	0.875	67.5	0.050	0.310	6.20
233.15	1518	0.883	69.2	0.051	0.279	5.40
243.15	1490	0.896	69.2	0.053	0.253	4.80
253.15	1460	0.909	70.9	0.054	0.235	4.40
263.15	1429	0.917	72.7	0.055	0.221	4.00
273.15	1397	0.934	72.7	0.056	0.213	3.80
283.15	1364	0.950	72.7	0.056	0.203	3.60
293.15	1330	0.963	72.7	0.056	0.198	3.50
303.15	1295	0.984	70.9	0.056	0.194	3.50
313.15	1257	1.005	69.2	0.056	0.191	3.50
323.15	1216	1.022	68.2	0.054	0.189	3.50
EUTECTIC CALCIUM CHLORIDE SOLUTION (29.9% CaCl₂)						
223.15	1320	2.608	401.2	0.117	36.352	312.00
233.15	1314	2.634	415.2	0.120	24.971	208.00
243.15	1310	2.663	429.2	0.123	17.177	139.00
253.15	1305	2.688	445.4	0.127	11.036	87.10
263.15	1300	2.713	458.2	0.130	6.956	53.60
273.15	1296	2.738	472.2	0.133	4.394	33.00
283.15	1291	2.763	485.0	0.136	3.353	24.60
293.15	1286	2.788	498.9	0.139	2.722	19.60
303.15	1282	2.813	510.6	0.142	2.267	16.00
313.15	1277	2.839	523.4	0.144	1.923	13.30
323.15	1272	2.868	535.0	0.147	1.654	11.30
ETHYLENE GLYCOL						
273.15	1131	2.294	241.9	0.093	57.523	615.00
293.15	1116	2.382	248.9	0.094	19.174	204.00
313.15	1101	2.474	255.9	0.094	8.686	93.00
333.15	1087	2.562	259.4	0.093	4.747	51.00
353.15	1077	2.650	261.7	0.092	2.982	32.40
373.15	1058	2.742	262.8	0.091	2.025	22.40

Continued..

Table A.8 PROPERTIES OF SELECTED SATURATED LIQUIDS

T	ρ	c_p	k	α	ν	p_r
K	kg/m ³	kJ/kg K	$\times 10^3$ W/m K	$\times 10^6$ m ² /s	$\times 10^6$ m ² /s	
ENGINE OIL (UNUSED)						
273.15	899	1.796	147.7	0.091	4282	47100
293.15	888	1.880	145.4	0.087	901	10400
313.15	876	1.964	144.2	0.083	241	2870
333.15	864	2.047	140.7	0.080	83	1050
353.15	852	2.131	138.4	0.077	37	490
373.15	840	2.219	137.2	0.074	20	276
393.15	828	2.307	134.9	0.071	12	175
413.15	816	2.395	133.8	0.069	7	116
433.15	805	2.483	131.4	0.066	5	84

TABLE A.9
PROPERTIES OF SELECTED GASES
 (At 1 Atmospheric Pressure)

T	ρ	c_p	k	α	μ	ν	p_r
K	kg/m ³	J/kg K	W/m K	$\times 10^5$ m ² /s	$\times 10^5$ kg/m s	$\times 10^5$ m ² /s	
AIR, Dry							
200	1.766	1003	0.0181	1.02	1.34	0.76	0.740
250	1.413	1003	0.0223	1.57	1.61	1.14	0.724
280	1.271	1004	0.0246	1.95	1.75	1.40	0.717
290	1.224	1005	0.0253	2.08	1.80	1.48	0.714
298	1.186	1005	0.0259	2.18	1.84	1.55	0.712
300	1.177	1005	0.0261	2.21	1.85	1.57	0.712
310	1.143	1006	0.0268	2.35	1.90	1.67	0.711
320	1.110	1006	0.0275	2.49	1.94	1.77	0.710
330	1.076	1007	0.0283	2.64	1.99	1.86	0.708
340	1.043	1007	0.0290	2.78	2.03	1.96	0.707
350	1.009	1008	0.0297	2.92	2.08	2.06	0.706
400	0.883	1013	0.0331	3.70	2.29	2.60	0.703
450	0.785	1020	0.0363	4.54	2.49	3.18	0.700
500	0.706	1029	0.0395	5.44	2.68	3.80	0.699
550	0.642	1039	0.0426	6.39	2.86	4.45	0.698
600	0.589	1051	0.0456	7.37	3.03	5.15	0.698
700	0.504	1075	0.0513	9.46	3.35	6.64	0.702
800	0.441	1099	0.0569	11.70	3.64	8.25	0.704
900	0.392	1120	0.0625	14.20	3.92	9.99	0.705
1000	0.353	1141	0.0672	16.70	4.18	11.8	0.709
1200	0.294	1175	0.0759	22.2	4.65	15.8	0.720
1400	0.252	1201	0.0835	27.6	5.09	20.2	0.732
1600	0.221	1240	0.0904	33.0	5.49	24.9	0.753
1800	0.196	1276	0.0970	38.3	5.87	29.9	0.772
2000	0.177	1327	0.1032	44.1	6.23	35.3	0.801
AMMONIA							
200	1.038	2199	0.0153	0.67	6.89	0.66	0.990
250	0.831	2248	0.0197	1.05	8.53	1.03	0.973
300	0.692	2298	0.0246	1.55	10.27	1.48	0.959
350	0.593	2349	0.0302	2.17	12.06	2.03	0.938
400	0.519	2402	0.0364	2.92	13.90	2.68	0.917
450	0.461	2455	0.0433	3.82	15.76	3.42	0.894
500	0.415	2507	0.0506	4.86	17.63	4.25	0.873
550	0.378	2559	0.0580	6.00	19.50	5.56	0.860
600	0.346	2611	0.0656	7.26	21.40	6.18	0.852
700	0.297	2710	0.0811	10.10	25.10	8.45	0.839
800	0.260	2810	0.0977	13.40	28.80	11.10	0.828

Continued..

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Table A.9 PROPERTIES OF SELECTED GASES
(At 1 Atmospheric Pressure)

T	ρ	c_p	k	α	μ	ν	Pr
K	kg/m ³	J/kg K	W/m K	$\times 10^5$ m ² /s	$\times 10^5$ kg/m s	$\times 10^5$ m ² /s	
ARGON							
200	2.435	523.6	0.0124	0.98	1.60	0.66	0.674
250	1.948	522.2	0.0152	1.49	1.95	1.00	0.672
300	1.623	521.6	0.0177	2.09	2.27	1.40	0.669
350	1.392	521.2	0.0201	2.78	2.57	1.85	0.666
400	1.218	521.0	0.0223	3.52	2.85	2.34	0.665
450	1.082	520.9	0.0244	4.33	3.12	2.88	0.665
500	0.974	520.8	0.0264	5.20	3.37	3.45	0.664
550	0.886	520.7	0.0283	6.14	3.60	4.07	0.662
600	0.812	520.6	0.0301	7.12	3.83	4.72	0.662
700	0.696	520.6	0.0336	9.28	4.25	6.11	0.658
800	0.609	520.5	0.0369	11.60	4.64	7.62	0.655
900	0.541	520.5	0.0398	14.10	5.01	9.26	0.654
1000	0.487	520.5	0.0427	16.80	5.35	11.00	0.652
1200	0.406	520.5	0.0481	22.80	5.99	14.80	0.648
1400	0.348	520.4	0.0535	29.60	6.56	18.90	0.638
CARBON DIOXIDE							
200	2.683	759	0.0095	0.47	1.02	0.38	0.814
250	2.146	806	0.0129	0.75	1.26	0.59	0.790
300	1.789	852	0.0166	1.09	1.50	0.84	0.768
350	1.533	897	0.0205	1.49	1.73	1.13	0.755
400	1.341	939	0.0244	1.94	1.94	1.45	0.747
450	1.192	979	0.0283	2.43	2.15	1.80	0.743
500	1.073	1017	0.0323	2.96	2.35	2.19	0.740
550	0.976	1049	0.0363	3.55	2.54	2.60	0.734
600	0.894	1077	0.0403	4.18	2.72	3.04	0.727
700	0.767	1126	0.0487	5.64	3.06	3.99	0.708
800	0.671	1169	0.0560	7.14	3.39	5.05	0.708
900	0.596	1205	0.0621	8.65	3.69	6.19	0.716
1000	0.537	1235	0.0680	10.25	3.97	7.40	0.721
1200	0.447	1283	0.0780	13.60	4.49	10.04	0.739
1400	0.383	1315	0.0867	17.20	4.97	13.00	0.754
CARBON MONOXIDE							
200	1.708	1045	0.0175	0.98	1.27	0.75	0.763
250	1.366	1048	0.0214	1.50	1.54	1.13	0.753
300	1.138	1051	0.0252	2.11	1.78	1.56	0.743
350	0.976	1056	0.0288	2.80	2.01	2.05	0.735
400	0.854	1060	0.0323	3.57	2.21	2.59	0.727
450	0.759	1065	0.0355	4.39	2.41	3.18	0.723
500	0.683	1071	0.0386	5.28	2.60	3.80	0.720
550	0.621	1077	0.0416	6.22	2.77	4.46	0.717
600	0.569	1084	0.0444	7.20	2.94	5.17	0.718

Continued..

Table A.9 PROPERTIES OF SELECTED GASES**(At 1 Atmospheric Pressure)**

T	ρ	c_p	k	α	μ	ν	p_r
K	kg/m ³	J/kg K	W/m K	$\times 10^5$ m ² /s	$\times 10^5$ kg/m s	$\times 10^5$ m ² /s	
700	0.488	1099	0.0497	9.270	3.25	6.66	0.718
800	0.427	1114	0.0549	11.500	3.54	8.29	0.718
900	0.379	1128	0.0596	13.900	3.81	10.04	0.721
1000	0.342	1142	0.0644	16.500	4.06	11.90	0.720
1100	0.310	1155	0.0692	19.300	4.30	13.9	0.718
1200	0.285	1168	0.0738	22.200	4.53	15.9	0.717
HELIUM							
200	0.2440	5197	0.1150	0.910	1.50	0.61	0.676
250	0.1952	5197	0.1340	1.540	1.75	0.90	0.680
300	0.1627	5197	0.1500	0.150	1.77	1.99	0.690
350	0.1394	5197	0.1650	0.165	2.28	2.21	0.698
400	0.1220	5197	0.1800	0.180	2.83	2.43	0.703
450	0.1085	5197	0.1950	0.195	3.45	2.63	0.702
500	0.0976	5197	0.2110	0.211	4.17	2.83	0.695
550	0.0887	5197	0.2290	0.229	4.97	3.02	0.684
600	0.0813	5197	0.2470	0.247	5.84	3.20	0.673
700	0.0697	5197	0.2780	0.278	7.67	3.55	0.663
800	0.0610	5197	0.3070	0.307	9.68	3.88	0.657
900	0.0542	5197	0.3350	0.335	11.90	4.20	0.652
1000	0.0488	5197	0.3630	0.363	14.30	4.50	0.645
1200	0.0407	5197	0.4160	0.416	19.70	5.08	0.635
1400	0.0349	5197	0.4690	0.469	25.90	5.61	0.622
1600	0.0305	5197	0.5210	0.521	32.90	6.10	0.608
1800	0.0271	5197	0.5700	0.570	40.40	6.57	0.599
2000	0.0244	5197	0.6200	0.620	48.90	7.00	0.587
HYDROGEN							
200	0.1299	13540	0.1280	0.770	0.68	0.55	0.717
250	0.0983	14070	0.1560	1.130	0.79	0.8	0.713
300	0.0819	14320	0.1820	1.550	0.89	1.09	0.705
350	0.0702	14420	0.2030	2.010	0.99	1.42	0.705
400	0.0614	14480	0.2210	2.490	1.09	1.78	0.714
450	0.0546	14500	0.2390	3.020	1.18	2.17	0.719
500	0.0492	14510	0.2560	3.590	1.27	2.59	0.721
550	0.0447	14520	0.2740	4.220	1.36	3.04	0.722
600	0.0410	14540	0.2910	4.890	1.45	3.54	0.724
700	0.0351	14610	0.3250	6.340	1.61	4.59	0.724
800	0.0307	14710	0.3600	7.970	1.77	5.76	0.723
900	0.0273	14840	0.3940	10.800	1.92	7.03	0.723
1000	0.0246	14990	0.4280	11.600	2.07	8.42	0.724
1200	0.0205	15370	0.4950	15.700	2.36	11.5	0.733
NITROGEN							
200	1.708	1043	0.0183	1.020	1.29	0.75	0.734

Continued..

Table A.9 PROPERTIES OF SELECTED GASES

(At 1 Atmospheric Pressure)

T	ρ	c_p	k	α	μ	ν	p_r
K	kg/m ³	J/kg K	W/m K	$\times 10^5$ m ² /s	$\times 10^5$ kg/m s	$\times 10^5$ m ² /s	
250	1.367	1042	0.0222	1.56	1.55	1.13	0.725
300	1.139	1040	0.0260	2.19	1.79	1.57	0.715
350	0.967	1041	0.0294	2.92	2.01	2.08	0.711
400	0.854	1045	0.0325	3.64	2.21	2.59	0.710
450	0.759	1050	0.0356	4.47	2.41	3.17	0.709
500	0.683	1057	0.0387	5.36	2.59	3.79	0.708
550	0.621	1065	0.0414	6.26	2.76	4.45	0.711
600	0.569	1075	0.0441	7.20	2.93	5.14	0.713
700	0.488	1098	0.0493	9.20	3.24	6.63	0.720
800	0.427	1122	0.0541	11.30	3.52	8.24	0.730
900	0.380	1146	0.0587	13.50	3.79	9.97	0.739
1000	0.342	1168	0.0631	15.80	4.04	11.80	0.747
1200	0.285	1205	0.0713	20.80	4.50	15.80	0.761
1400	0.244	1233	0.0797	26.50	4.92	20.20	0.761
OXYGEN							
200	1.951	906	0.0182	1.03	1.47	0.75	0.728
250	1.561	914	0.0225	1.58	1.78	1.14	0.721
300	1.301	920	0.0267	2.23	2.07	1.59	0.711
350	1.115	929	0.0306	2.95	2.34	2.10	0.710
400	0.976	942	0.0342	3.72	2.59	2.65	0.713
450	0.867	956	0.0377	4.55	2.83	3.26	0.717
500	0.780	971	0.0412	5.44	3.05	3.91	0.720
550	0.709	987	0.0447	6.38	3.27	4.61	0.722
600	0.650	1003	0.0480	7.36	3.47	5.34	0.725
700	0.557	1032	0.0544	9.46	3.85	6.91	0.730
800	0.488	1054	0.0603	11.70	4.21	8.63	0.736
900	0.434	1074	0.0661	14.20	4.54	10.50	0.738
1000	0.390	1091	0.0717	16.80	4.85	12.40	0.738
1200	0.325	1116	0.0821	22.60	5.42	16.70	0.737
1400	0.278	1136	0.0921	29.10	5.95	21.30	0.734
WATER VAPOUR (STEAM)							
300	0.0253*	2041	0.0181	35.1*	0.91	36.1*	1.030
350	0.258*	2037	0.0222	4.22*	1.12	4.33*	1.020
400	0.555	2000	0.0264	2.38	1.32	2.38	1.000
450	0.491	1968	0.0307	3.17	1.52	3.10	0.980
500	0.441	1977	0.0357	4.09	1.73	3.92	0.960
550	0.401	1994	0.0411	5.15	1.93	4.82	0.940
600	0.367	2022	0.0464	6.25	2.13	5.82	0.930
700	0.314	2083	0.0572	8.74	2.54	8.09	0.930
800	0.275	2148	0.0686	11.6	2.95	10.70	0.920
900	0.244	2217	0.0780	14.4	3.36	13.70	0.950
1000	0.22	2288	0.0870	17.3	3.76	17.10	0.990

* at Saturation Pressure (less than 1 atm)

TABLE A.10
EMISSION VALUES OF SELECTED METALS AND NON METALS

A. METALS

Material	T K	ϵ
Aluminum		
Polished	300 - 900	0.04 - 0.06
Commercial sheet	400	0.09
Heavily oxidized	400 - 800	0.2 - 0.3
Anodized	300	0.8
Bismuth		
Bright	350	0.34
Brass		
Highly polished	500 - 650	0.03 - 0.04
Polished	350	0.09
Dull plate	300 - 600	0.22
Oxidized	450 - 800	0.6
Chromium		
Polished	300 - 1400	0.08 - 0.4
Copper		
Commercial sheet	300	0.15
Highly polished	300	0.02
Polished	300-500	0.04 - 0.05
Oxidized	600-1000	0.5-0.8
Black Oxidized	300	0.78
Gold		
Highly polished	300 - 1000	0.03 - 0.06
Bright foil	300	0.07
Iron		
Highly polished	300 - 500	0.05 - 0.07
Case iron	300	0.44
Wrought Iron	300 - 500	0.28
Rusted	300	0.61
Oxidized	300 - 900	0.64 - 0.78
Lead		
Polished	300 - 500	0.06 - 0.08
Unoxidized, rough	300	0.43
Oxidized	300	0.63
Magnesium		
Polished	300 - 500	0.07 - 0.13
Mercury		
	300 - 400	0.09 - 0.12
Molybdenum		
Polished	300 - 2000	0.05 - 0.21
Oxidized	600 - 800	0.8 - 0.82
Nickel		
Polished	500 - 1200	0.07 - 0.17
Oxidized	450 - 1000	0.37 - 0.57
Platinum		
Polished	500 - 1500	0.06 - 0.18
Silver		
Polished	300 - 1000	0.02 - 0.07
Stainless steel		
Polished	300 - 1000	0.17 - 0.3
Lightly oxidized	600 - 1000	0.3 - 0.4
Heavily oxidized	600 - 1000	0.7 - 0.8
Steel		
Polished sheet	300 - 500	0.08 - 0.14
Commercial sheet	500 - 1200	0.2 - 0.32
Heavily oxidized	300	0.81
Tin		
Polished	300	0.05
Tungsten		
Polished	300 - 2500	0.03 - 0.29
Filament	3500	0.39
Zinc		
Polished	300 - 800	0.02 - 0.05
Oxidized	300	0.25

Continued..

Table A.10 EMISSIVITY VALUES OF SELECTED METALS AND NON METALS

B. NON METALS

Material	T	ε
	K	
Alumina	800 - 1400	0.65 - 0.45
Aluminum oxide	600 - 1500	0.69 - 0.41
Asbestos	300	0.96
Asphalt	300	0.85 - 0.93
Brick:		
Red (common)	300	0.93 - 0.96
Fireclay	1200	0.75
Carbon filament	2000	0.53
Cloth	300	0.75 - 0.9
Concrete	300	0.88 - 0.94
Glass:		
Window	300	0.9 - 0.95
Pyrex	300 - 1200	0.82 - 0.62
Pyroceram	300 - 1500	0.85 - 0.57
Ice	273	0.95 - 0.99
Magnesium oxide	400 - 800	0.69 - 0.55
Masonry	300	0.80
Paints:		
Aluminum	300	0.4 - 0.5
Lacquer, shiny, black	300	0.82
Oil, all colors	300	0.92 - 0.96
Red primer	300	0.93
Acrylic enamel white	300	0.90
Acrylic enamel black	300	0.876

[illegible]

SECTION B**CONDUCTION HEAT TRANSFER****CONTENTS**

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B.1 STEADY STATE CONDUCTION

B.1.1 One Dimensional Heat Transfer in Extended Surfaces

(a) Relations

Assumptions: Fins of finite length and loses heat by convection from its end.

$$\frac{T - T_{\infty}}{T_o - T_{\infty}} = \frac{\cosh m(L - x) + (h/mk) \sinh m(L - x)}{\cosh mL + (h/mk) \sinh mL}$$

$$q = \sqrt{hPkA}(T_o - T_{\infty}) \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL}$$

(b) Fin Efficiency

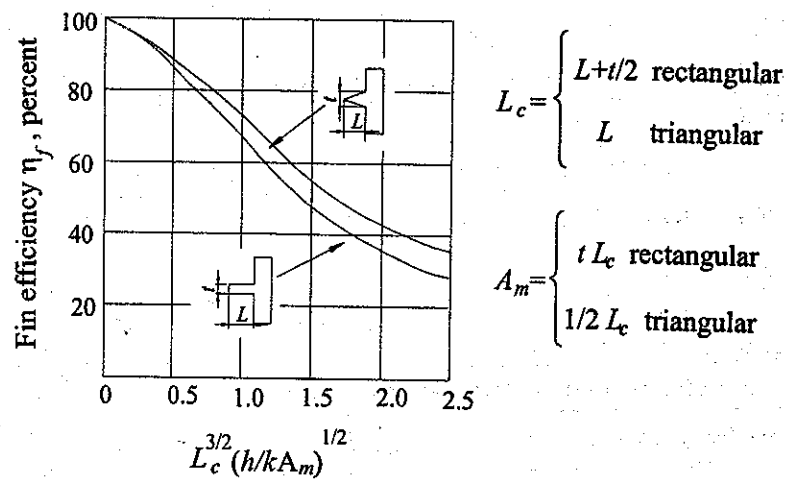


Fig. B.1.1 Efficiency of rectangular and triangular fins.

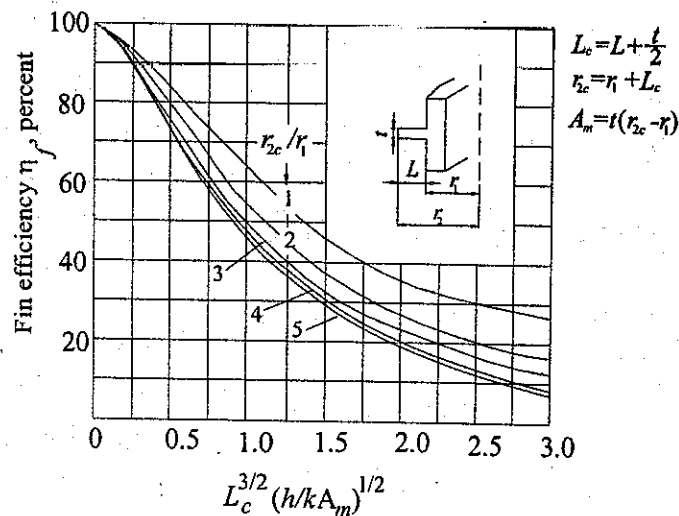


Fig. B.1.2 Efficiency of circumferential fins of rectangular profile.

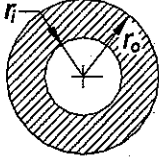
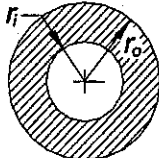
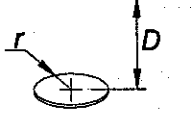
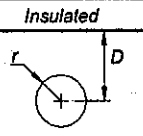
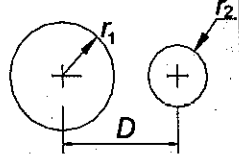
B.1.2 Multi-Dimensional Heat Conduction

TABLE B.1.1
CONDUCTION SHAPE FACTOR RELATIONS

Physical system	Schematic	Shape factor	Remarks
Isothermal cylinder of radius r and length L , buried in semi-infinite medium having isothermal surface		$\frac{2\pi L}{\cosh^{-1}(D/r)}$ $\frac{2\pi L}{\ln(D/r)}$	$L \gg r$ $L \gg r$ $D > 3r$
Isothermal sphere of radius r , buried in infinite medium		$4\pi r$	
Isothermal sphere of radius r , buried in semi-infinite medium having isothermal surface. $\Delta T = T_{\text{surf}} - T_{\text{far field}}$		$\frac{4\pi r}{1 - r/2D}$	
Conduction between two isothermal cylinders of length L buried in infinite medium. r_1, r_2 are the radii of bigger and smaller cylinders respectively		$\frac{2\pi L}{\cosh^{-1}\left(\frac{D^2 - r_1^2 - r_2^2}{2r_1 r_2}\right)}$	$L \gg r$ $L \gg D$
Buried cube in infinite medium, L on a side		$8.24 L$	
Isothermal cylinder of length L , radius r placed in semi-infinite medium as shown.		$\frac{2\pi L}{\ln(2L/r)}$	$L \gg 2r$
Isothermal rectangular parallelepiped buried in semi infinite medium having isothermal surface		$1.685L \left[\log\left(1 + \frac{b}{a}\right) \right]^{-0.59} \left(\frac{b}{c}\right)^{-0.078}$	
Plane wall, thickness L and area A as shown		$\frac{A}{L}$	One dimensional heat flow

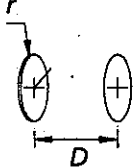
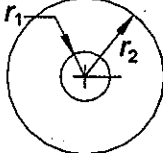
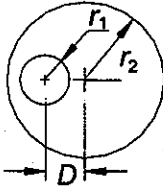
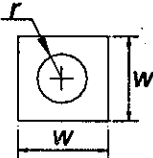
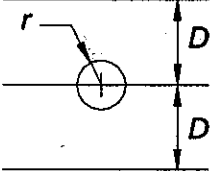
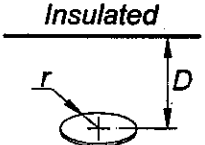
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Table B.1.1 CONDUCTION SHAPE FACTOR RELATIONS

Physical system	Schematic	Shape factor	Remarks
Hollow cylinder, length L inner radius r_i and outer radius r_o		$\frac{2\pi L}{\ln\left(\frac{r_o}{r_i}\right)}$	$L \gg r$
Hollow sphere inner radius r_i and outer radius r_o		$\frac{4\pi r_o r_i}{r_o - r_i}$	
Thin horizontal disk buried in semi-infinite medium with isothermal surface. r is radius of disc; D is the distance between surface of medium and disc.		$4r$	$D = 0$
		$8r$	$D \gg 2r$
		$\frac{4\pi r}{(\pi/2) - \tan^{-1}(r/2D)}$	$\frac{D}{2r} \gg 1$ $\tan^{-1}\left(\frac{r}{2D}\right)$ in radians
Hemisphere, radius r , buried in semi infinite medium $\Delta T = T_{\text{sphere}} - T_{\text{far field}}$		$2\pi r$	
Isothermal sphere, radius r , buried in semi infinite medium with insulated surface		$\frac{4\pi r}{1 + r/2D}$	
Two isothermal spheres buried in infinite medium. r_1, r_2 are the radii of bigger and smaller spheres respectively		$\frac{4\pi r_2}{\frac{r_2}{r_1} \left[1 - \frac{(r_1/D)^4}{1 - (r_2/D)^2} \right] - \frac{2r_2}{D}}$	$D \gg 5 r_{\text{max}}$
Thin rectangular plate of length L , buried in semi- infinite medium		$\frac{\pi w}{\ln(4w/L)}$	$D = 0$ $w > L$
		$\frac{2\pi w}{\ln(4w/L)}$	$D \gg w$ $w > L$
		$\frac{2\pi w}{\ln(2\pi D/L)}$	$w \gg L$ $D > 2w$

Continued..

Table B.1.1 CONDUCTION SHAPE FACTOR RELATIONS

Physical system	Schematic	Shape factor	Remarks
Parallel discs buried in infinite medium. r is the radius of the discs; D is the distance between the discs		$\frac{4\pi r}{\left[\frac{\pi}{2} - \tan^{-1}(r/D)\right]}$	$D > 5r$ $\tan^{-1}\left(\frac{r}{D}\right)$ in radians
Concentric cylinders of length L ; r_1 and r_2 are the radii of smaller and bigger cylinders.		$\frac{2\pi}{\ln\left(\frac{r_2}{r_1}\right)}$	
Eccentric cylinders of length L ; r_1 and r_2 are the radii of smaller and bigger cylinders; D is the eccentricity.		$\frac{2\pi L}{\cosh^{-1}\left(\frac{r_1^2 + r_2^2 - D^2}{2r_1 r_2}\right)}$	$L \gg r_2$
Cylinder centered in square tube of length L ; r is the radius of the cylinder		$\frac{2\pi L}{\ln(0.54w/r)}$	$L \gg w$
Horizontal cylinder of length L centered in infinite plates (isothermal) r is the radius of cylinder		$\frac{2\pi L}{\ln(4D/r)}$	
Thin horizontal disk of radius r buried in semi infinite medium with adiabatic surface $\Delta T = T_{disc} - T_{far field}$		$\frac{4\pi r}{(\pi/2) + \tan^{-1}(r/2D)}$	$D/2r \gg 1$ $\tan^{-1}\left(\frac{r}{2D}\right)$ in radians

B.2 TRANSIENT HEAT CONDUCTION

TABLE B.2.1
THE ERROR FUNCTION

x	$\text{erf}(x)$	x	$\text{erf}(x)$	x	$\text{erf}(x)$
0.00	0.00000	0.80	0.74210	1.60	0.976360
0.02	0.02256	0.82	0.75381	1.62	0.978040
0.04	0.04511	0.84	0.76514	1.64	0.979620
0.06	0.06762	0.86	0.77610	1.66	0.981100
0.08	0.09008	0.88	0.78669	1.68	0.982490
0.10	0.11246	0.90	0.79691	1.70	0.983790
0.12	0.13476	0.92	0.80677	1.72	0.985000
0.14	0.15695	0.94	0.81627	1.74	0.986130
0.16	0.17901	0.96	0.82542	1.76	0.987190
0.18	0.20094	0.98	0.83423	1.78	0.9881700
0.20	0.22270	1.00	0.84270	1.80	0.989090
0.22	0.24430	1.02	0.85084	1.82	0.989940
0.24	0.26570	1.04	0.85865	1.84	0.990740
0.26	0.28690	1.06	0.86614	1.86	0.991470
0.28	0.30788	1.08	0.87333	1.88	0.992160
0.30	0.32863	1.10	0.88020	1.90	0.992790
0.32	0.34913	1.12	0.88079	1.92	0.993380
0.34	0.36936	1.14	0.89308	1.94	0.993920
0.36	0.38933	1.16	0.89910	1.96	0.994430
0.38	0.40901	1.18	0.90484	1.98	0.994890
0.40	0.42839	1.20	0.91031	2.00	0.995322
0.42	0.44749	1.22	0.91553	2.10	0.997020
0.44	0.46622	1.24	0.92050	2.20	0.998137
0.46	0.48466	1.26	0.92524	2.30	0.998857
0.48	0.50275	1.28	0.92973	2.40	0.999311
0.50	0.52050	1.30	0.93401	2.50	0.999593
0.52	0.53790	1.32	0.93806	2.60	0.999764
0.54	0.55494	1.34	0.94191	2.70	0.999866
0.56	0.57162	1.36	0.94556	2.80	0.999925
0.58	0.58792	1.38	0.94902	2.90	0.999959
0.60	0.60386	1.40	0.95228	3.00	0.999978
0.62	0.61941	1.42	0.95538	3.20	0.999994
0.64	0.63459	1.44	0.95830	3.40	0.999998
0.66	0.64938	1.46	0.96105	3.60	1.000000
0.68	0.66278	1.48	0.96365		
0.70	0.67780	1.50	0.96610		
0.72	0.69143	1.52	0.96841		
0.74	0.70468	1.54	0.97059		
0.76	0.71754	1.56	0.97263		
0.78	0.73001	1.58	0.97455		

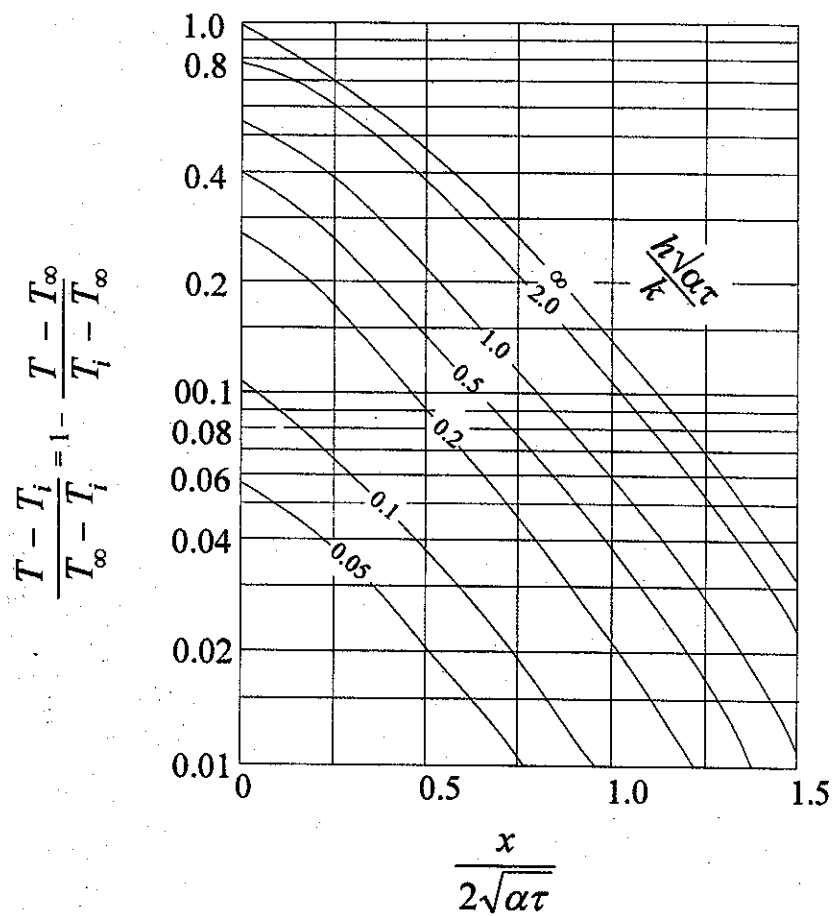


Fig. B.2.1 Temperature distribution in the semi-infinite solid with convection boundary condition

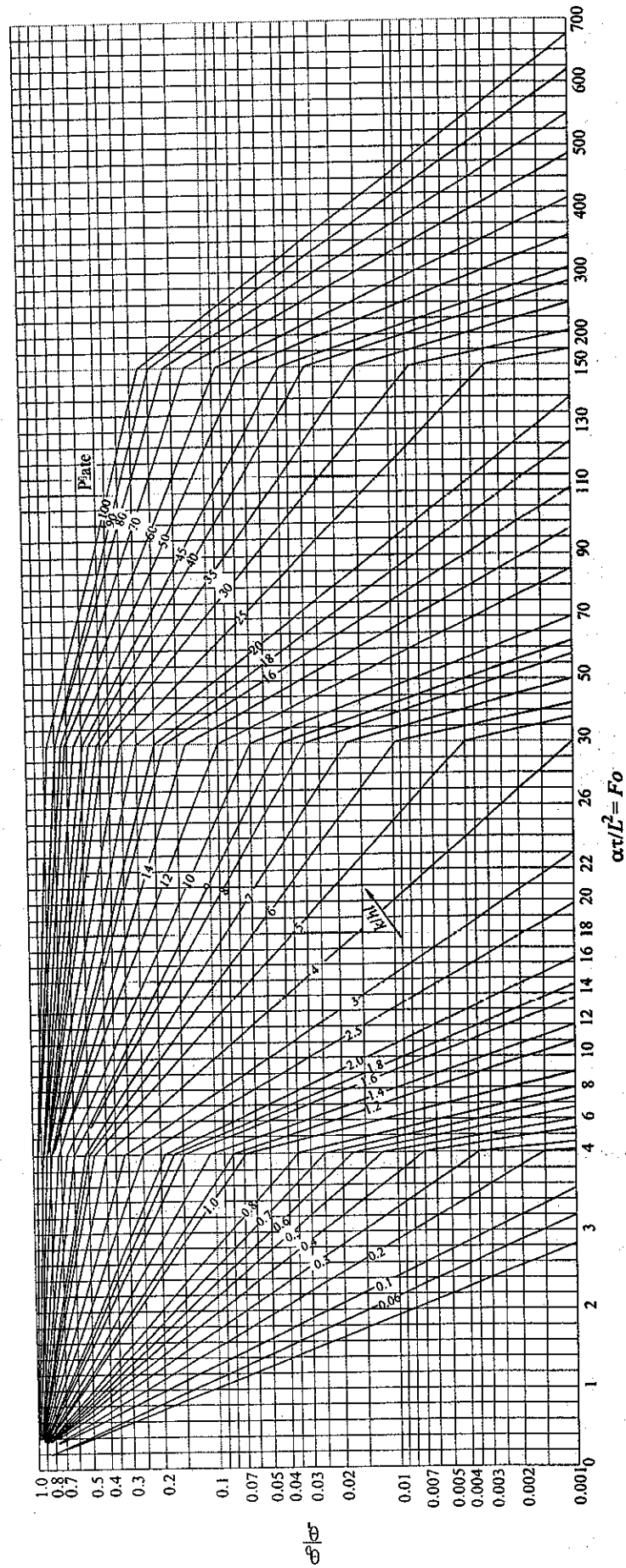


Fig. B.2.2 Midplane temperature for an infinite plate of thickness $2L$.

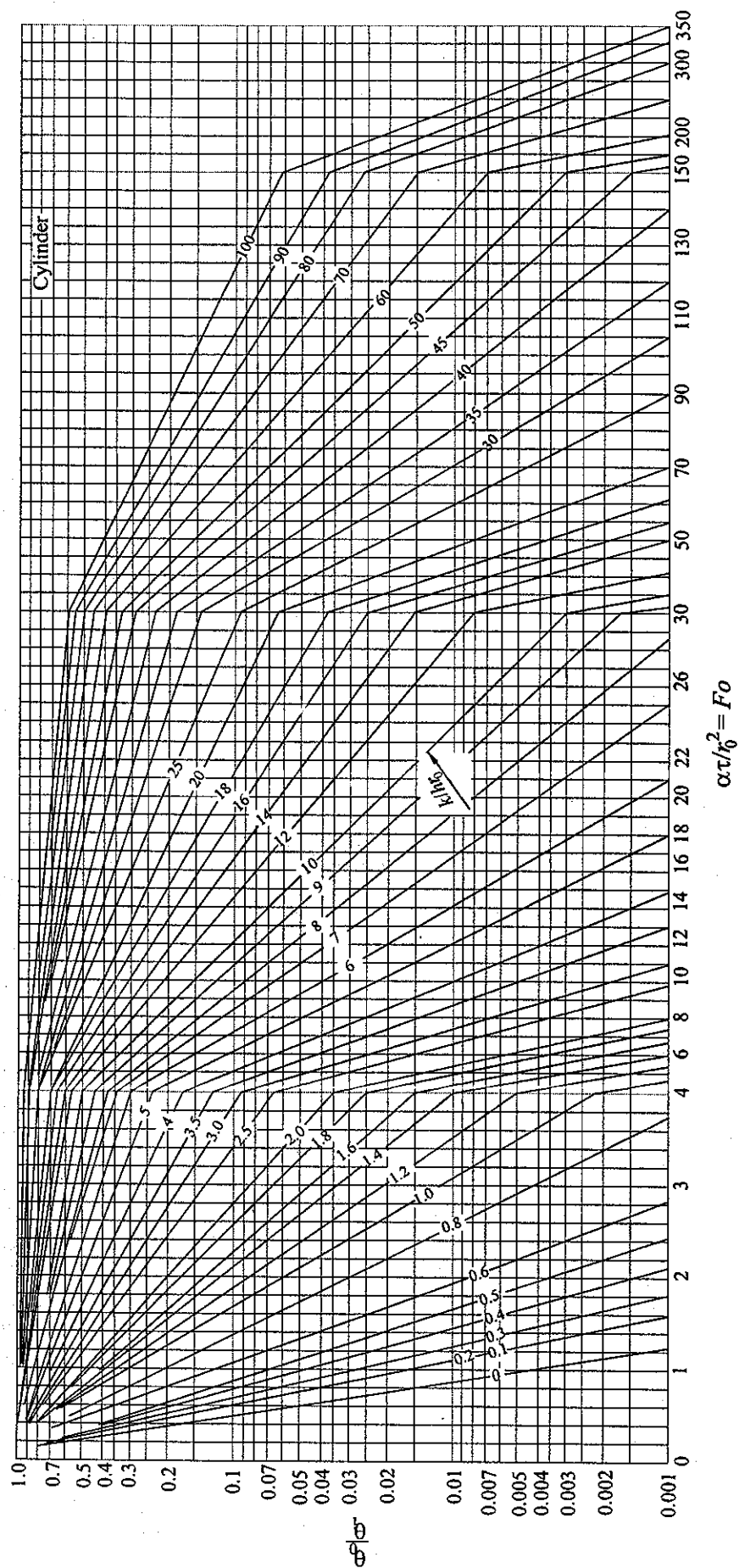


Fig. B.2.3 Axis temperature for an infinite cylinder of radius r_0 .

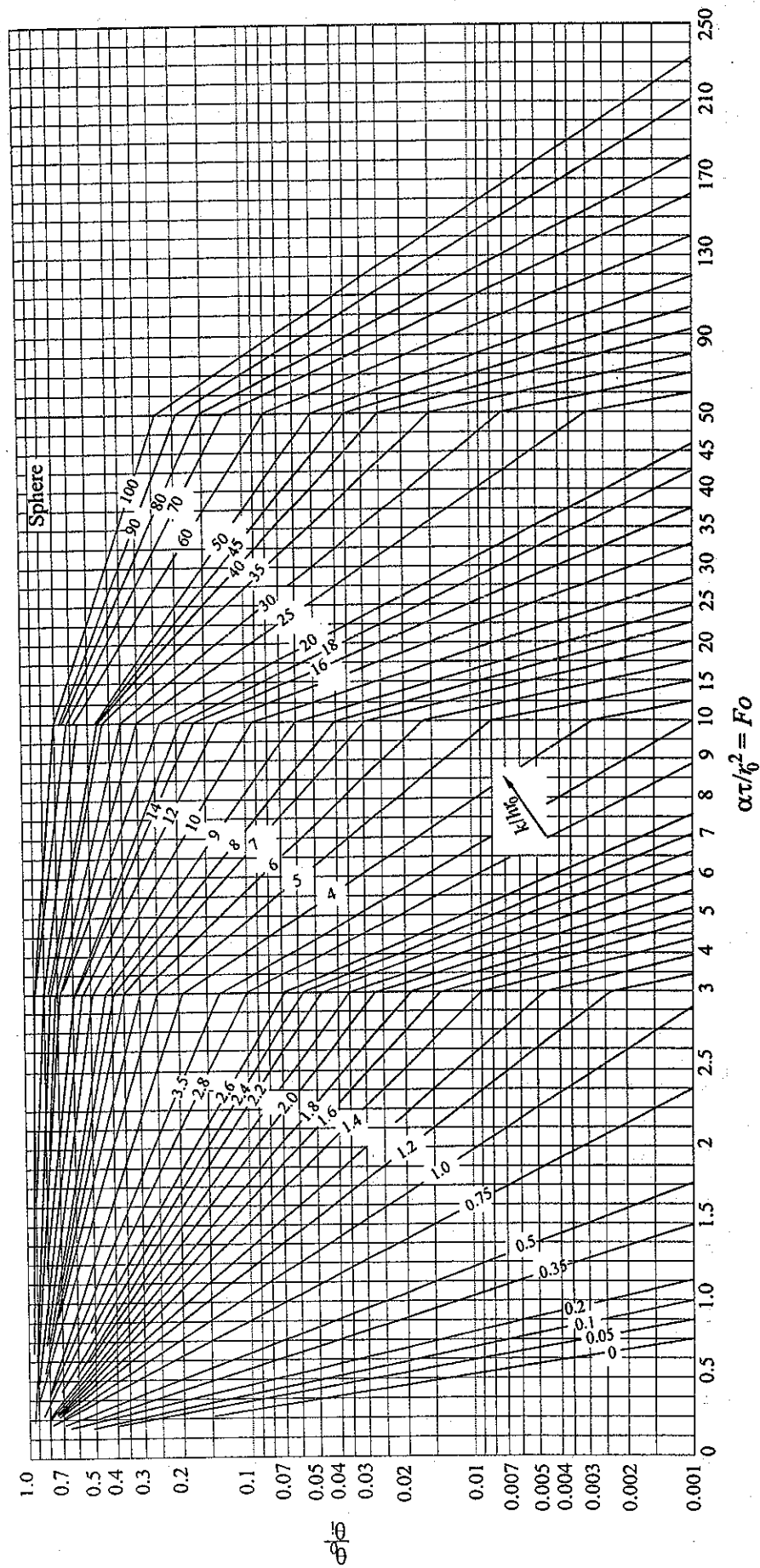


Fig. B.2.4 Center temperature for a sphere of radius r_0 .

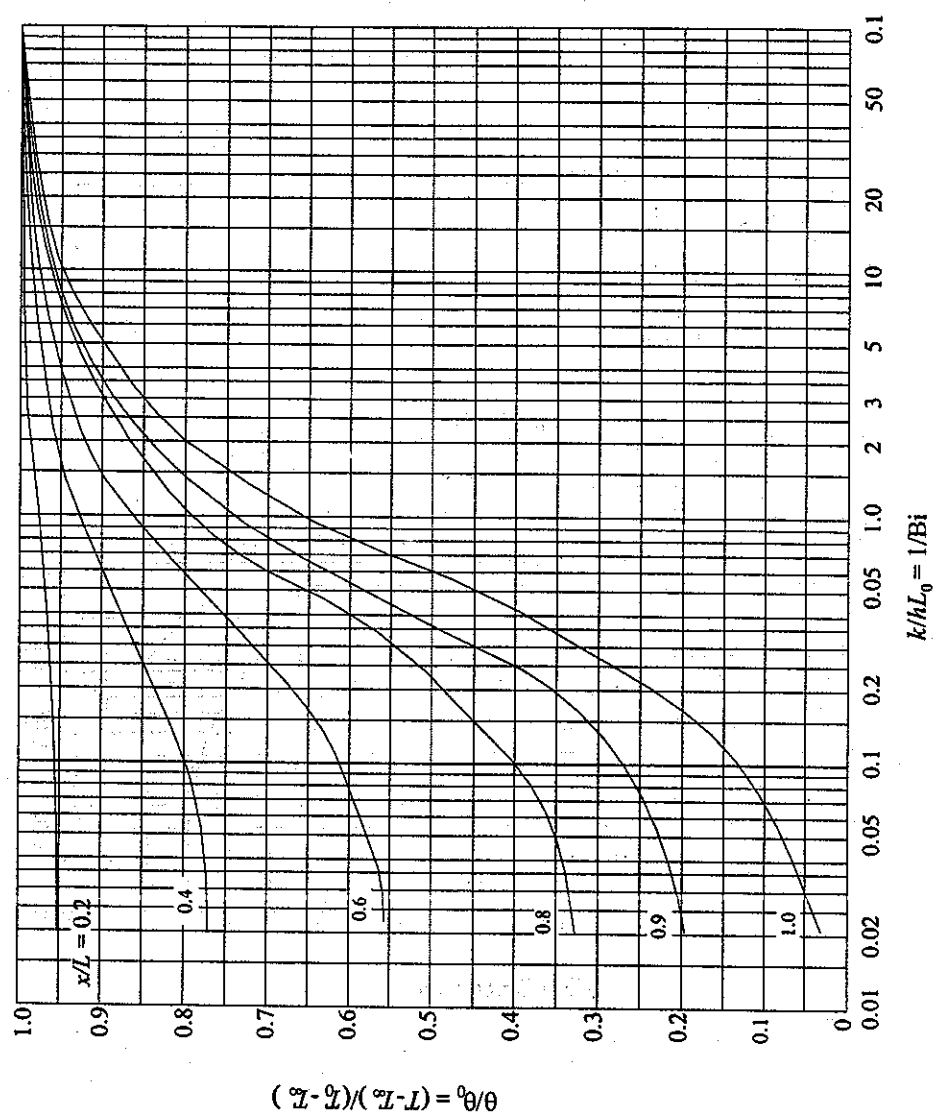


Fig. B.2.5. Temperature as a function of center temperature in an infinite plate of thickness $2L$.

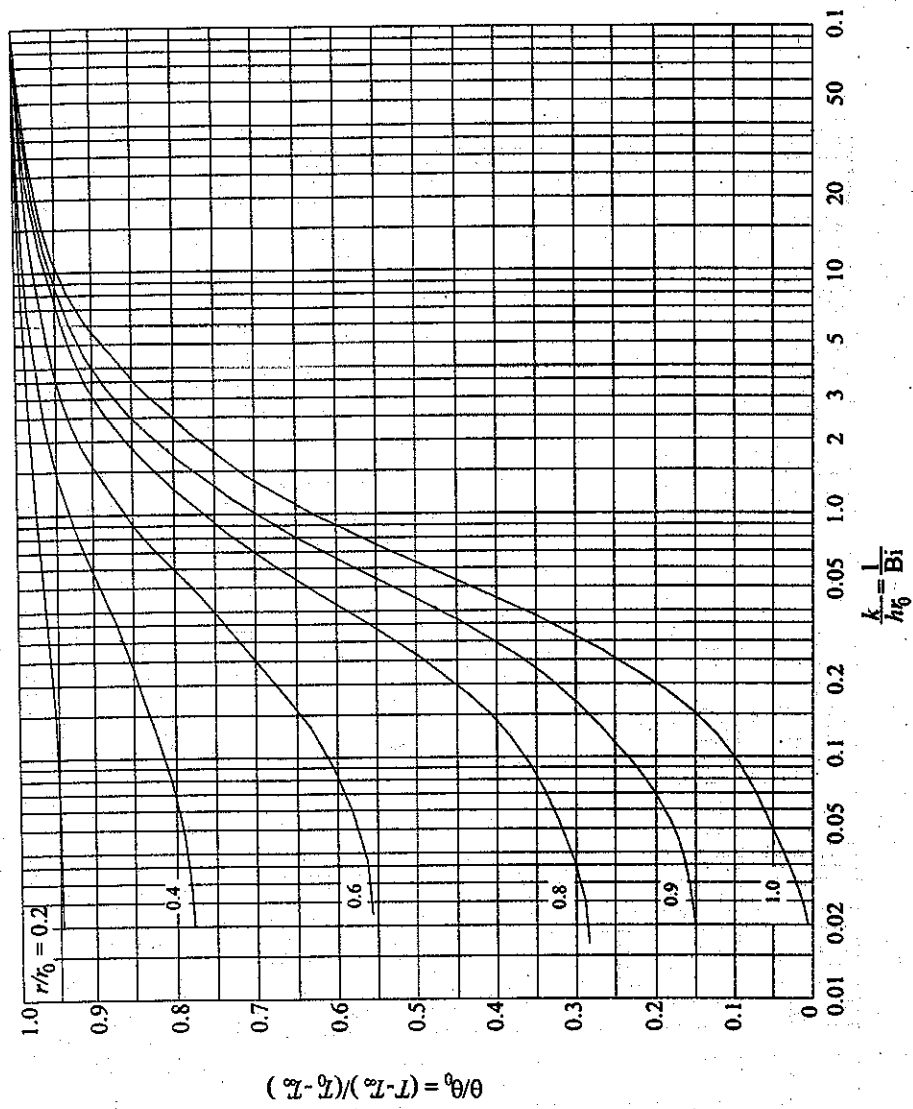


Fig. B.2.6 Temperature as a function of axis temperature in an infinite cylinder of radius r_0 .

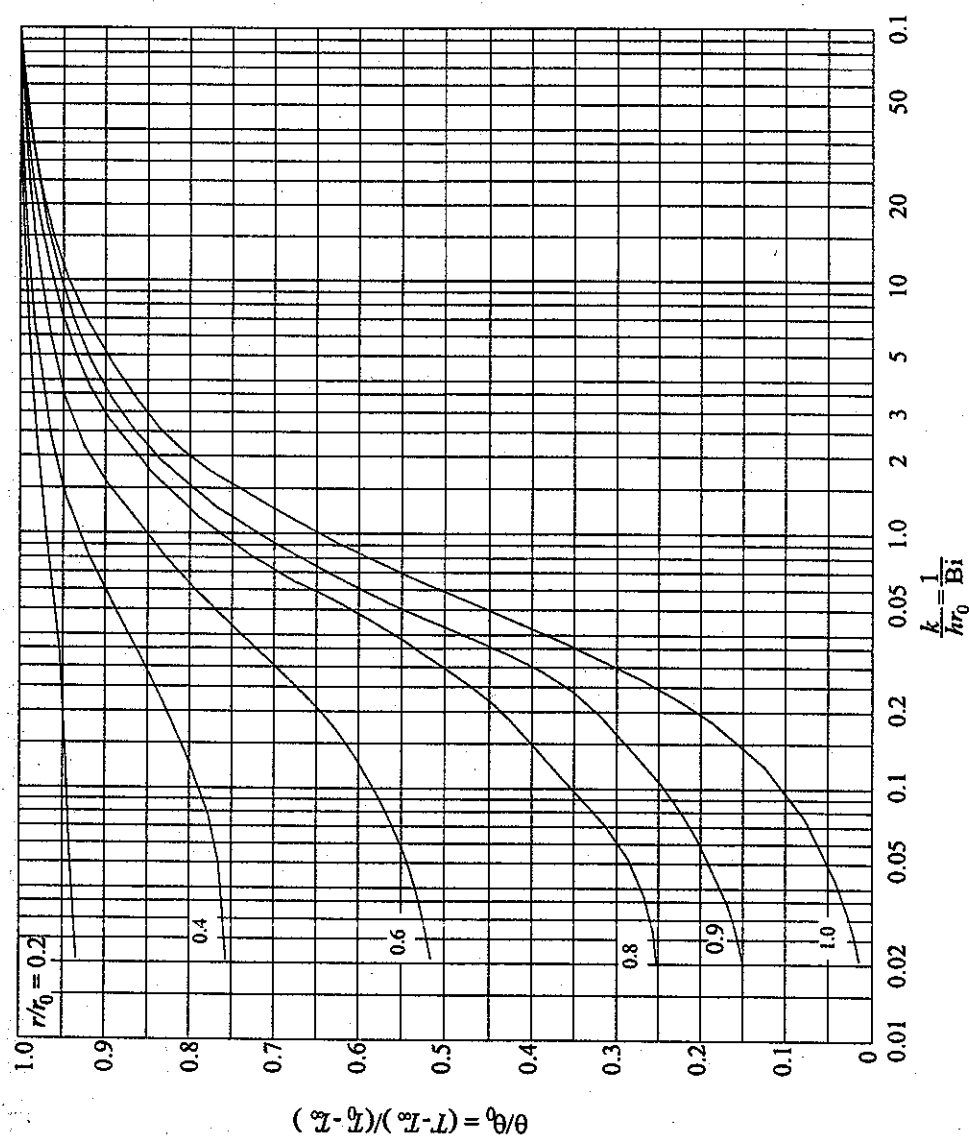


Fig. B.2.7 Temperature as a function of center temperature for a sphere of radius r_0 .

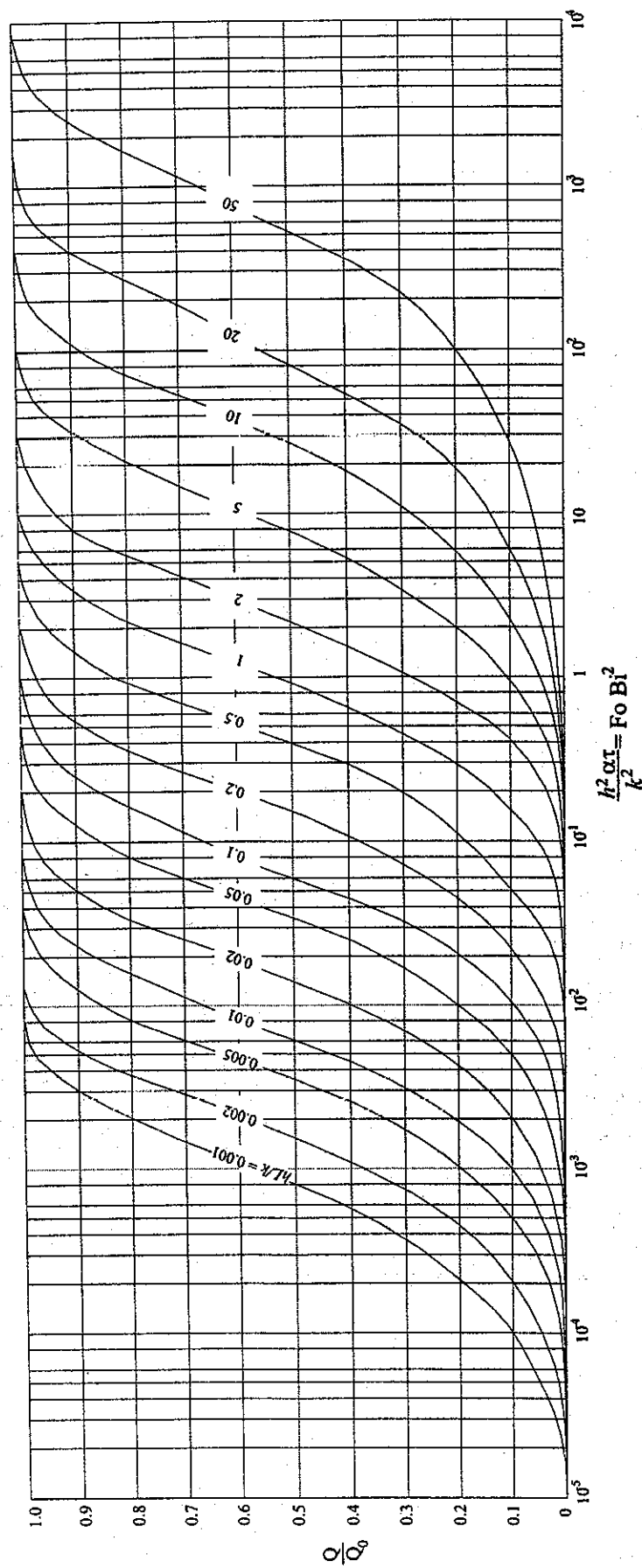


Fig. B.2.8 Dimensionless heat loss Q/Q_0 of an infinite plane of thickness $2L$ with time.

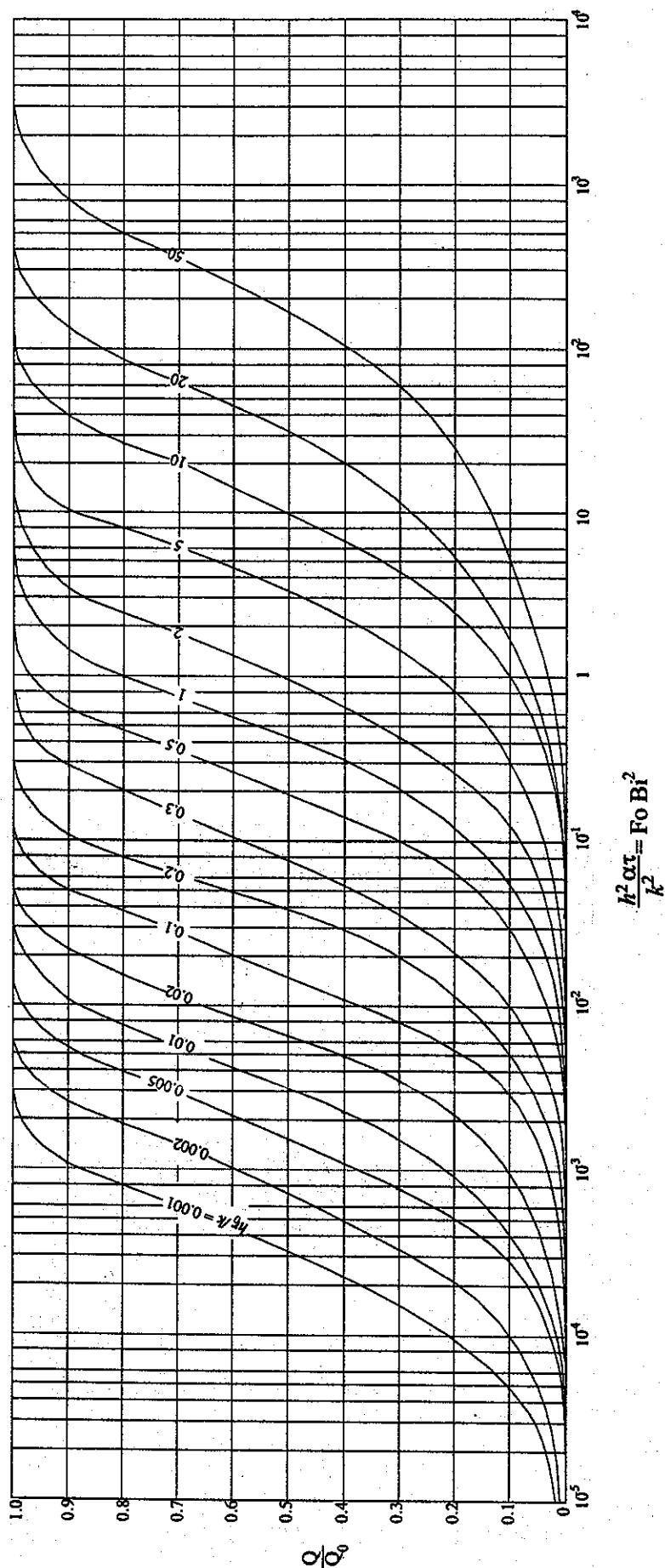


Fig. B.2.9 Dimensionless heat loss Q/Q_0 of an infinite cylinder of radius r_0 with time.

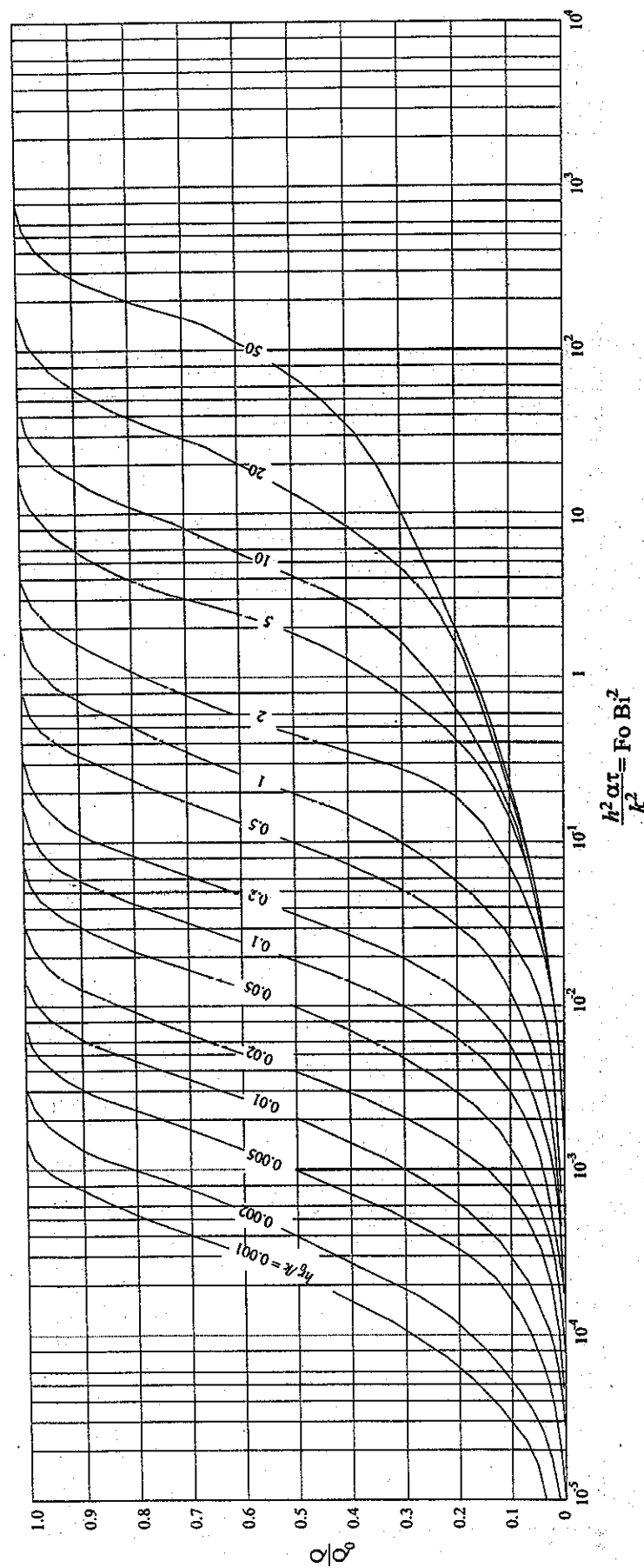


Fig. B.2.10 Dimensionless heat loss Q/Q_0 of a sphere of radius r_0 with time

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C.1 FORCED CONVECTION RELATIONS

TABLE C.1.1

FORCED CONVECTION RELATIONS FOR FLOW OVER FLAT PLATE
(Nu_x is Nusselt number at any x)

FLOW CONDITIONS	CORRELATION	REMARKS
Laminar Flow		
Constant wall temperature	$Nu_x = 0.332 Re_x^{0.5} Pr^{0.333}$	$0.6 < Pr < 50$
If heating starts from " x_0 " from leading edge	$Nu_x = 0.332 Re_x^{0.5} Pr^{0.333} \left[1 - \left(\frac{x_0}{x} \right)^{0.75} \right]^{-0.33}$	$T_w = \text{constant}$
For liquid metals (low Prandtl numbers) and for silicones (high Prandtl numbers)	$Nu_x = \frac{0.3387 Re_x^{0.5} Pr^{0.333}}{[1 + (0.0468/Pr)^{0.67}]^{0.25}}$	$Pr < 0.05$; $Re_x Pr > 100$; $Pr > 50$ $T_w = \text{constant}$
For liquid metals only	$Nu_x = 0.565 [Re_x Pr]^{0.5}$	$Pr \leq 0.05$
Constant heat flux	$Nu_x = 0.453 Re_x^{0.5} Pr^{0.333}$ $T_w - T_\infty = \frac{(q_l/k)}{[0.6795 Re_x^{0.5} Pr^{0.333}]}$	$0.6 < Pr < 50$
For liquid metals and silicones constant heat flux	$Nu_x = \frac{0.453 Re_x^{0.5} Pr^{0.333}}{[1 + (0.0207/Pr)^{0.67}]^{0.25}}$	Valid for $Pr > 50$ and $Pr \leq 0.05$ and for $Re_x Pr > 100$
Average values of Nu_x for constant heat fluxes or constant wall temperature	$\overline{Nu} = 2 Nu_{at x=1}$	
Turbulent Flow ($5 \times 10^5 < Re_x < 10^7$)		
Hydrodynamic boundary layer thickness at x	$\delta_{hx} = 0.381 Re_x^{-0.2}$	
Fully turbulent from leading edge	$Nu_x = 0.0296 Re_x^{0.8} Pr^{0.333}$	
Constant heat flux	$Nu_x = 1.04$ times Nu_x calculated as per constant wall temperature.	

* Drag coefficient correlation for laminar flow

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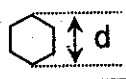
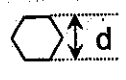
$$Nu = 0.036 Re_x^{0.8} Pr^{0.333}$$

Table C.1.1 FORCED CONVECTION RELATIONS FOR FLOW OVER FLAT PLATE
(Nu_x is Nusselt number at any x)

FLOW CONDITIONS	CORRELATION	REMARKS
Laminar-Turbulent Combined Flow		
Local and average skin friction coefficient	$\delta_{hx} = 0.381 Re_x^{-0.2} - 0.10256 Re_x^{-1}$ $C_{fx} = 0.0592 Re_x^{-0.2};$ $(500000 > Re_x > 10^7)$ $C_{fx} = 0.37 [\log_{10} Re_x]^{-2.364}$ $C_{fl} = 0.074 Re_x^{-0.2} - 1742 Re_x^{-1}$ $(\text{with critical } Re \text{ as } 5 \times 10^5)$	C_{fx} is local skin friction coefficient. C_{fl} is average skin friction coefficient up to length l .
Constant wall temperature	$\overline{Nu} = Pr^{0.333} [0.037 Re_L^{0.8} - 871]$ $Re_L \leq 10^8; 0.6 < Pr < 60$ or generally $\overline{Nu} = Pr^{0.333} [0.037 Re_L^{0.8} - A]$ where $A = 0.037 Re_{cr}^{0.8} - 0.664$ $Re_{cr}^{0.5}$	$\overline{Nu}$ is average nusselts number up to a length L . Re_{cr} is critical Reynolds number

* Fluid properties are evaluated at the film temperature $T_f = \frac{(T_w - T_\infty)}{2}$

TABLE C.1.2
FORCED CONVECTION RELATIONS FOR FLOW ACROSS CYLINDERS,
SPHERES

FLOW CONDITIONS	CORRELATION AND ASSOCIATED CONSTANTS				REMARKS
Flow across cylinders	$Nu_f = C Re_{df}^n Pr^{0.333}$				$0.4 < Re_{df} < 400000$
	Re_{df}	C		n	
	0.4 - 4	0.9890		0.330	
	4 - 40	0.9110		0.385	
	40 - 4×10^3	0.6830		0.466	
	4×10^3 - 4×10^4	0.1930		0.618	
	4×10^4 - 4×10^5	0.0266		0.805	
Flow across non-circular cylinders	$Nu_f = C Re_{df}^n Pr^{0.333}$				
	Geometry	Re_{df}	C	n	
	 "d" diagonal	$5 \times 10^3 - 10^5$	0.246	0.588	
	 "d" side	$5 \times 10^3 - 10^5$	0.102	0.675	
		$5 \times 10^3 - 1.95 \times 10^4$	0.160	0.638	
		$1.95 \times 10^4 - 10^5$	0.0385	0.782	
		$5 \times 10^3 - 10^5$	0.153	0.638	
	 "d" side	$4 \times 10^3 - 1.5 \times 10^4$	0.228	0.731	
Flow across spheres	$Nu_f = 0.037 Re_{df}^{0.8}$				Pr approx 0.7 (gases); $17 < Re < 70000$
	$Nu Pr^{-0.3} (\mu_w / \mu)^{0.25} = 1.2 + 0.53 Re_d^{0.54}$				Water and oils $1 < Re < 200000$, properties at T_a
	$Nu = 2 + (0.4 Re_d^{0.5} + 0.06 Re_d^{0.67}) Pr^{0.4} (\mu_a / \mu_w)^{0.25}$				$3.5 < Re < 80000$ $0.7 < Pr < 380$;

Subscripts: b : bulk temperature; f : film temperature; ∞ : free stream temperature;
 w : wall temperature.

TABLE C.1.3
FORCED CONVECTION RELATIONS FOR FLOW ACROSS TUBE BANKS

Correlations and Associated Constants										
$N \geq 10$	$Nu_f = C Re_{f \max}^n Pr_f^{0.333}$									
$S_n/d >$	$S_n/d = 1.25$		$S_n/d = 1.5$		$S_n/d = 2.0$		$S_n/d = 3.0$			
S_p/d	C	n	C	n	C	n	C	n		
In - line	1.25	0.386	0.592	0.305	0.608	0.111	0.704	0.0703	0.752	
	1.5	0.407	0.586	0.278	0.62	0.112	0.702	0.0753	0.744	
	2.0	0.464	0.57	0.332	0.602	0.254	0.632	0.22	0.648	
	3.0	0.322	0.601	0.396	0.584	0.415	0.581	0.317	0.608	
	Staggered									
0.6	---	---	---	---	---	---	0.236	0.636		
0.9	---	---	---	---	0.495	0.571	0.445	0.581		
1.0	---	---	0.552	0.558	---	---	---	---		
1.125	---	---	---	---	0.531	0.565	0.575	0.560		
1.25	0.575	0.556	0.561	0.554	0.576	0.556	0.579	0.562		
1.5	0.501	0.568	0.511	0.562	0.502	0.568	0.542	0.568		
2.0	0.448	0.572	0.462	0.568	0.535	0.556	0.498	0.570		
3.0	0.344	0.592	0.395	0.58	0.488	0.562	0.467	0.574		
Ratio of 'h' for 'N' rows deep to that for 10 rows deep										
N	1	2	3	4	5	6	7	8	9	10
Ratio for staggered arrangement	0.68	0.75	0.83	0.89	0.92	0.95	0.97	0.98	0.99	1.0
Ratio for inline arrangement	0.64	0.8	0.87	0.9	0.92	0.94	0.96	0.98	0.99	1.0
$N \geq 20$	$Nu = C Re_{d \max}^n Pr_f^{0.333} (Pr/Pr_w)^{0.25}$									
Geometry	$Re_{d \max}$			C			n			
In - line	10 - 100			0.8			0.4			
	100 - 1000			Treat as individual tubes						
	$10^3 - 2 \times 10^5$			0.27			0.63			
	$> 2 \times 10^5$			0.9			0.4			
Staggered	10 - 100			0.9			0.4			
	100 - 1000			Treat as individual tubes						
	$10^3 - 2 \times 10^5$			$0.35(s_n/s_l)^{0.2}$ for $(s_n/s_l) < 2$			0.6			
	$10^3 - 2 \times 10^5$			0.4 for $(s_n/s_l) > 2$			0.6			
	$> 2 \times 10^5$			0.022			0.84			
Ratio of 'h' for 'N' rows deep to that for 20 rows deep										
N	2	3	4	5	6	8	10	16	20	---
Ratio for staggered arrangement	0.77	0.84	0.89	0.92	0.94	0.97	0.98	0.99	1	---
Ratio for inline arrangement	0.7	0.8	0.9	0.92	0.94	0.97	0.98	0.99	1	---

Subscripts: *b*: bulk temperature; *f*: film temperature; ∞ : free stream temperature; *w*: wall temperature.

S_n : Transverse pitch; S_p : Longitudinal pitch

TABLE C.1.4
FORCED CONVECTION RELATIONS FOR INTERNAL FLOWS

FLOW CONDITIONS	CORRELATION	REMARKS
Tube flow, fully developed turbulent flow,	$Nu_d = 0.023 Re_d^{0.8} Pr^n$	$n = 0.4$ for heating $n = 0.3$ for cooling $0.6 < Pr < 100$; $2500 < Re < 1.25 \times 10^5$
Tube flow	$Nu_d = 0.0214(Re_d^{0.8} - 100)Pr^{0.4}$	$0.5 < Pr < 1.5$; $10^4 < Re < 5 \times 10^6$
	$Nu_d = 0.012(Re_d^{0.87} - 280)Pr^{0.4}$	$1.5 < Pr < 500$; $3000 < Re < 10^6$
Tube flow, fully developed turbulent flow,	$Nu_d = 0.027 Re_d^{0.8} Pr^{0.67} (\mu/\mu_w)^{0.14}$	
Tube flow, entrance region	$Nu_d = 0.036 Re_d^{0.8} Pr^{0.33} (d/L)^{0.55}$	
Tube flow	$Nu_d = \frac{(f/8) Re_d Pr}{1.07 + 12.7(f/8)^{0.5} (Pr^{0.67} - 1)} \left(\frac{\mu_b}{\mu_w} \right)^n$ Where $f = (1.82 \log_{10} Re_d - 1.64)^{-2}$	$0.5 < Pr < 200$ for 6% accuracy $0.5 < Pr < 2000$ for 10% accuracy $10^4 < Re_d < 5 \times 10^6$ $0.8 < \left(\frac{\mu_b}{\mu_w} \right)^n < 40$
Tube flow, laminar flow	$Nu_d = 3.66 + \frac{0.0668(d/L) Re_d Pr}{1 + 0.04[(d/L) Re_d Pr]^{0.67}}$	
Tube flow, fully developed laminar flow	$Nu_d = 1.86(Re_d Pr)^{0.33} (d/L)^{0.33} (\mu/\mu_w)^{0.14}$	$Re_d Pr(d/L) > 10$
Rough tubes, fully developed turbulent flow	$St_b Pr_f^{0.67} = f/8$	
Non-circular tubes	All the above equations can be used at the hydraulic mean diameter.	

Subscripts: b : bulk temperature; f : film temperature; ∞ : free stream temperature; w : wall temperature.

C.2 NATURAL CONVECTION RELATIONS

TABLE C.2.1
NATURAL CONVECTION FROM EXTERNAL SURFACES

CONDITIONS/ GEOMETRY	RELATION	REMARKS		
A variety of isothermal surfaces		Values of C and m		
Vertical planes and Cylinders	$\overline{Nu}_f = C(Gr_f Pr_f)^m$	$Gr_f Pr_f$	C	m
		$10^4 - 10^9$	0.590	0.250
		$10^9 - 10^{13}$	0.100	0.333
Horizontal cylinders		$10^4 - 10^9$	0.530	0.250
		$10^9 - 10^{12}$	0.130	0.333
		$10^{-10} - 10^{-2}$	0.675	0.058
		$10^{-2} - 10^2$	1.020	0.148
		$10^2 - 10^4$	0.850	0.188
		$10^4 - 10^7$	0.480	0.250
		$10^7 - 10^{12}$	0.125	0.333
Upper surface of heated plates or lower surface of cooled plates		$2 \times 10^4 - 8 \times 10^6$	0.540	0.250
Upper surface of heated plates or lower surface of cooled plates		$8 \times 10^6 - 10^{11}$	0.150	0.333
Lower surface of heated plates or upper surface of cooled plates		$10^5 - 10^{11}$	0.270	0.250
Vertical cylinder, height = diameter, Characteristic length is diameter		$10^4 - 10^6$	0.775	0.210
Irregular solids, Characteristic length is distance fluid particle travels in boundary layer		$10^4 - 10^9$	0.520	0.250
Vertical isothermal surface	$\overline{Nu}^{0.5} = 0.825 + \left[\frac{0.387 Ra^{0.167}}{[1 + (0.492 / Pr)^{0.5625}]^{0.296}} \right]$	$10^{-1} < Re_L < 10^{12}$		
Vertical surface, constant heat flux, local h Gr_f^* is modified Grashof number Gr_x	$Nu_{xf} = C(Gr_f^* Pr_f)^m$	C=0.6,m=0.2 for $10^5 < Gr_f^* < 10^{11}$		
		C = 0.17,m = 0.25 for $2 \times 10^{13} < Gr_f^* < 10^{16}$		

Continued..

Table C.2.1 NATURAL CONVECTION FROM EXTERNAL SURFACES

CONDITIONS/ GEOMETRY	RELATION	REMARKS										
Isothermal horizontal cylinders	$\overline{Nu}^{0.5} = 0.6 + 0.387 \left[\frac{GrPr}{1 + (0.559/Pr)^{0.5625}} \right]^{0.167}$	$10^{-5} < GrPr < 10^{13}$										
Horizontal Surface, constant heat flux		All properties except β are evaluated at $T_e = T_w - 0.25(T_w - T_a)$										
For heated surface facing upward	$\overline{Nu}_l = 0.13(Gr_l Pr)^{0.333}$ $\overline{Nu}_l = 0.16(Gr_l Pr)^{0.333}$	For $Gr_l Pr < 2 \times 10^8$ For $2 \times 10^8 < Gr_l Pr < 10^{11}$										
For heated surface facing downward	$\overline{Nu}_l = 0.58(Gr_l Pr)^{0.2}$	For $10^6 < Gr_l Pr < 10^{11}$										
Inclined plates		θ is the angle with which the plate makes with the vertical. A positive θ indicates the heated surface faces downward.										
with heated surface facing downward with constant heat flux	$\overline{Nu}_e = 0.56(Gr_e Pr_e \cos \theta)^{0.25}$	$\theta < 88^\circ$; $10^5 < Gr_e Pr_e \cos \theta < 10^{11}$ All properties except β are evaluated at $T_e = T_w - 0.25(T_w - T_a)$ β is evaluated at $T_a + 0.5(T_w - T_a)$										
For almost horizontal plates	$\overline{Nu}_e = 0.58(Gr_e Pr_e \cos \theta)^{0.2}$	$88^\circ < \theta < 90^\circ$; $10^6 < Gr_e Pr_e < 10^{11}$										
with heated surface facing upward	$\overline{Nu}_e = 0.14[(Gr_e Pr_e)^{0.333} - (Gr_e Pr_e)^{0.333}] + 0.56(Gr_e Pr_e \cos \theta)^{0.25}$	$10^5 < Gr_e Pr_e \cos \theta < 10^{11}$; Gr_e is critical Grashof number <table><tr><th>θ</th><th>Gr_e</th></tr><tr><td>-15</td><td>5×10^9</td></tr><tr><td>-30</td><td>2×10^9</td></tr><tr><td>-60</td><td>10^8</td></tr><tr><td>-75</td><td>10^6</td></tr></table>	θ	Gr_e	-15	5×10^9	-30	2×10^9	-60	10^8	-75	10^6
θ	Gr_e											
-15	5×10^9											
-30	2×10^9											
-60	10^8											
-75	10^6											
Spheres	$Nu = 2 + 0.43(GrPr)^{0.25}$ $Nu = 2 + 0.5(GrPr)^{0.25}$ $Nu = 2 + \left[\frac{0.589(GrPr)^{0.25}}{1 + (0.469/Pr)^{0.5625}} \right]^{0.4444}$	$1 < GrPr < 10^5$ Water, $3 \times 10^5 < GrPr < 8 \times 10^8$ $0.5 < Pr$ $GrPr < 10^{11}$										

TABLE C.2.2

NATURAL CONVECTION IN ENCLOSED SPACES

The basic equation is $q = k_e A(\Delta T / \delta)$ in which k_e is calculated using the equation $\frac{k_e}{k} = C(Gr_\delta Pr)^n (L/\delta)^m$. Constants C , m , and n are obtained from the following table.

FLUID	GEOMETRY	$Gr_\delta Pr$	Pr	L/δ	C	n	m
Gas	Vertical Plate, isothermal	<2000	---	---	1	0	0
		6000 - 2×10^5	0.5 - 2	11-42	0.197	0.25	-0.1111
		2×10^5 - 1.1×10^7	0.5 - 2	11-42	0.073	0.333	-0.1111
	Horizontal plate, isothermal heated from below	<1700	---	---	1	0	0
		1700 - 7000	0.5 - 2	---	0.059	0.4	0
		7000 - 3.2×10^5	0.5 - 2	---	0.212	0.25	0
		$>3.2 \times 10^5$	0.5 - 2	---	0.061	0.333	0
Liquid	Vertical plate, constant heat flux or isothermal	<2000	---	---	1	0	0
		10^4 - 10^7	$1 - 2 \times 10^4$	10 - 40	$Nu_\delta = 0.42(Gr_\delta Pr)^{0.25} Pr^{0.012} \left(\frac{L}{\delta}\right)^{-0.3}$		
		10^6 - 10^9	1 - 20	1 - 40	$Nu_\delta = 0.046(Gr_\delta Pr)^{0.333}$		
	Horizontal plate, isothermal, Heated from below	<1700	---	---	1	0	0
		1700 - 6000	1 - 5000	---	0.012	0.6	0
		6000 - 37000	1 - 5000	---	0.375	0.2	0
		37000 - 10^8	1 - 20	---	0.13	0.3	0
		$>10^8$	1 - 20	---	0.057	0.33333	0
	Vertical annulus	Same as vertical plate					
	Horizontal annulus, isothermal	$6000 - 10^6$	1 - 5000	---	0.11	0.29	0
		$10^6 - 10^8$	1 - 5000	---	0.4	0.2	0
	Spherical annulus	$120 - 1.1 \times 10^9$	0.7-4000	--	0.228	0.226	0

(continued from page 51)

1. 100% of the total

2. 100% of the total

3. 100% of the total

4. 100% of the total

5. 100% of the total

6. 100% of the total

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30. 100% of the total

31. 100% of the total

32. 100% of the total

33. 100% of the total

34. 100% of the total

SECTION D**BOILING AND CONDENSATION HEAT TRANSFER****CONTENTS**

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D.1 RELATIONS FOR CONDENSATION HEAT TRANSFER

(a) Vertical plate (laminar film condensation)

The film thickness, $\delta = \left[\frac{4\mu kx(T_g - T_w)}{gh_{fg}\rho_l(\rho_l - \rho_v)} \right]^{1/4}$

Local heat transfer coefficient, $h_x = \left[\frac{\rho_l(\rho_l - \rho_v)gh_{fg}k^3}{4\mu x(T_g - T_w)} \right]^{1/4}$

Local Nusselt number, $Nu_x = \left[\frac{\rho_l(\rho_l - \rho_v)gh_{fg}x^3}{4k\mu(T_g - T_w)} \right]^{1/4}$

Average heat transfer coefficient, $\bar{h} = 0.943 \left[\frac{\rho_l(\rho_l - \rho_v)gh_{fg}k^3}{L\mu_f(T_g - T_w)} \right]^{1/4}$

Film Reynolds number, $Re_f = \left[\frac{4\bar{h}L(T_{sat} - T_w)}{h_{fg}\mu_f} \right]$

(b) Outside horizontal tubes (laminar film condensation)

Average heat transfer coefficient, $\bar{h} = 0.725 \left[\frac{\rho_l(\rho_l - \rho_v)gh_{fg}k_f^3}{\mu_f d(T_g - T_w)} \right]^{1/4}$

(c) Inclined surfaces/cylinders

If the plate/cylinder is inclined at an angle " Φ " with horizontal, replace the gravitational force with its component parallel to the heat transfer surface, or $g' = g \sin \Phi$.

(d) Inside horizontal tubes (laminar film condensation of refrigerants)

For low vapor flow rates ($Re_m < 35,000$):

$$\bar{h} = 0.555 \left[\frac{\rho_l(\rho_l - \rho_v)gk^3 h'_{fg}}{\mu d(T_g - T_w)} \right]^{1/4}$$

For higher flow rates:

$$\frac{\bar{h}d}{k_f} = 0.026 Pr_f^{1/3} Re_m^{0.8}$$

where, Re_m is mixture Reynolds number given by

$$Re_m = \frac{d}{\mu_f} \left[G_f + G_v \left(\frac{\rho_f}{\rho_v} \right)^{1/2} \right]$$

Subscripts l and v denote liquid and vapor, respectively

(e) The condensation number

$$Co = \bar{h} \left[\frac{\mu^2}{k^3 \rho_l (\rho_l - \rho_v) g} \right]^{1/3}$$

$$\text{or } Co = C^{4/3} \left(\frac{4 \sin \phi (A/P)}{L} \right)^{1/3} Re_f^{-1/3}$$

For vertical plate:

$$Co = \frac{1.47}{Re_f^{1/3}} \quad \text{for } Re_f < 1800$$

For horizontal cylinder:

$$Co = \frac{1.514}{Re_f^{1/3}} \quad \text{for } Re_f < 1800$$

When turbulence is encountered: $Co = 0.0077 Re_f^{0.4}$ for $Re_f > 1800$

D.2 RELATIONS FOR BOILING HEAT TRANSFER**(a) Nucleate pool boiling**

$$\frac{C_l \Delta T_x}{h_{fg} Pr_l^s} = \left[C_{sf} \left(\frac{\frac{q}{A}}{\mu_l h_{fg}} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \right) \right]^{0.333}$$

where,

C_l = Specific heat of saturated liquid, J/kg K

ΔT_x = Excess temperature, K

h_{fg} = Enthalpy of vaporization, J/kg

Pr_l = Prandtl number of saturated liquid

q/A = heat flux per unit area, W/m² K

μ_l = Liquid viscosity, kg/m s

σ = Surface tension of liquid-vapor interface, N/m (See Table D.2.1)

g = gravitational acceleration, m/s²

ρ_l = Density of saturated liquid, kg/m³

ρ_v = Density of saturated vapor, kg/m³

C_{sf} = Constant, determined from experimental data (See Table D.2.1)

$s = 1.0$ for water and 1.7 for other liquids

TABLE D.2.1
VAPOR-LIQUID SURFACE TENSION FOR WATER AND VALUE OF THE
COEFFICIENT C_{sf} FOR VARIOUS LIQUID-SURFACE COMBINATION

Saturation temperature of water $^{\circ}\text{C}$	Surface tension of water σ , mN/m	Fluid-heating-surface combination	C_{sf}
0.00	75.60	Water-copper	0.01300
15.56	73.30	Water-platinum	0.01300
37.78	69.80	Water-brass	0.00600
60.00	66.00	Water-emery-polished-copper	0.01280
93.33	60.10	Water-ground and polished stainless steel	0.00800
100.00	58.80	Water-chemically etched stainless steel	0.01330
160.00	46.10	Water-mechanically polished stainless steel	0.01320
226.67	32.00	Water-emery polished and paraffin-treated copper	0.01470
293.33	16.20	Water-scored copper	0.00680
360.00	1.46	Water-Teflon pitted stainless steel	0.00580
374.10	0.00	Carbon tetrachloride-copper	0.01300
		Carbon tetrachloride-emery polished copper	0.00700
		Benzene-chromium	0.01000
		n-butyl alcohol-copper	0.00305
		Ethyl alcohol-chromium	0.02700
		Isopropyl alcohol-copper	0.00225
		n-pentane-chromium	0.01500
		n-pentane-emery polished copper	0.01540
		n-pentane-emery polished nickel	0.01270
		n-pentane-lapped copper	0.00490
		n-pentane-emery-rubbed copper	0.00740
		35% K_2CO_3 -copper	0.00540
		50% K_2CO_3 -copper	0.00270

(i) For low pressure boiling water:

$$q/A = 2.253(\Delta T_x)^{3.96} \text{ W/m}^2 \quad \text{For } 0.7 < p < 14 \text{ MPa}$$

(ii) For higher pressure:

$$(q/A) = 283.2 p^{4/3} (\Delta T_x)^3 \text{ W/m}^2 \quad \text{For } 0.7 < p < 14 \text{ Mpa}$$

(iii) For peak heat flux in nucleate boiling:

$$(q/A)_{\max} = \frac{\pi}{24} h_{fg} \rho_v \left[\frac{\sigma g (\rho_l - \rho_v)}{\rho_v^2} \right]^{1/4} \left(1 + \frac{\rho_v}{\rho_l} \right)^{1/2}$$

$$\text{Also } \frac{Q_{\max}}{\rho_L d^3 \lambda} = 1.83 \times 10^{-3} \left(\frac{\rho_l^2 V^2 d}{\rho_v \sigma g_c} \right)^{0.341}$$

where,

$Q_{\max}$ = Maximum heat transfer per drop

ρ_L = Density of liquid droplet

V = normal component of impact velocity

ρ_v = Vapor density evaluated at film temperature

σ = Surface tension

d = drop diameter

λ = Modified heat of vaporization, defined by $\lambda = h_{fg} + c_{pv} \left(\frac{T_w - T_{sat}}{2} \right)$

(b) Peak boiling heat flux on horizontal cylinder

$$\frac{q''_{\max}}{q''_{\max F}} = 0.89 + 2.27 \exp(-3.44 \sqrt{R'}) \quad \text{For } 0.15 < R'$$

$$\text{where, } R' = R \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2}$$

(c) Peak heat flux on infinite horizontal plate

$$q''_{\max F} = 0.131 \sqrt{\rho_v} h_{fg} [\sigma g (\rho_l - \rho_v)]^{1/4}$$

(d) Heat transfer coefficient in the stable film-boiling region on a horizontal tube

$$h_b = 0.62 \left[\frac{k_g^3 \rho_v (\rho_l - \rho_v) g (h_{fg} + 0.4 c_{pv} \Delta T_x)}{d \mu_v \Delta T_x} \right]^{1/4}$$

(e) The total heat transfer coefficient with convection and radiation effects in to consideration

$$h = h_b \left(\frac{h_b}{h} \right)^{1/3} + h_r$$

where, h_r is radiative heat transfer coefficient defined by

$$h_r = \frac{\sigma \varepsilon (T_w^4 - T_{sat}^4)}{T_w - T_{sat}}$$

where, σ is Stefan-Boltzmann constant and ε is the emissivity of the surface

TABLE D.2.2

SIMPLIFIED RELATIONS FOR BOILING HEAT-TRANSFER FOR WATER
AT ATMOSPHERIC PRESSURE

$$(\Delta T_x = T_w - T_{sat}, ^\circ\text{C})$$

Surface	q/A KW/m ²	H W/m ² K	Approximate range of °C	Approximate range of h W/m ² K
Horizontal	$q/A < 16$	$1042(\Delta T_x)^{1/3}$	0-7.76	0-2060
	$16 < q/A < 240$	$5.56(\Delta T_x)^{1/3}$	7.32-14.4	2180-6600
Vertical	$q/A < 3$	$537(\Delta T_x)^{1/7}$	0-4.51	0-670
	$3 < q/A < 63$	$7.96(\Delta T_x)^{1/3}$	4.41-9.43	680-680

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TABLE E.1
RADIATION SHAPE FACTOR RELATIONS

SNO.	GEOMETRY	SHAPE FACTOR
1.	Parallel, equal rectangles (Fig. E.1) $x = \frac{X}{D}, y = \frac{Y}{D}$	$F_{1-2} = \left(\frac{2}{\pi xy} \right) \left\{ \ln \left[\frac{(1+x^2)(1+y^2)}{(1+x^2+y^2)} \right]^{1/2} + x(1+y^2)^{1/2} \tan^{-1} \left[\frac{x}{(1+y^2)^{1/2}} \right] + y(1+x^2)^{1/2} \tan^{-1} \left[\frac{y}{(1+x^2)^{1/2}} \right] - x \tan^{-1} x - y \tan^{-1} y \right\}$
2.	Parallel, equal, coaxial disks (Fig. E.2) $R = \frac{d}{2x}, X = \frac{(2R^2+1)}{R^2}$	$F_{1-2} = \frac{[X - (X^2 - 4)^{1/2}]}{2}$
3.	Perpendicular rectangle with a common edge (Fig. E.3) $H = \frac{Z}{X}, W = \frac{Y}{X}$	$F_{1-2} = \left(\frac{1}{\pi W} \right) \left\{ W \tan^{-1} \left(\frac{1}{W} \right) + H \tan^{-1} \left(\frac{1}{H} \right) - (H^2 + W^2)^{1/2} \tan^{-1} \left[\frac{1}{H^2 + W^2} \right] + \left(\frac{1}{4} \right) \ln \left[\frac{(1+W^2)(1+H^2)}{(1+W^2+H^2)} \right] \times \left[\frac{W^2(1+W^2+H^2)}{(1+W^2)(W^2+H^2)} \right]^{W^2} \times \left[\frac{H^2(1+H^2+W^2)}{(1+H^2)(H^2+W^2)} \right]^{H^2} \right\}$

Continued.

Table E.1 RADIATION SHAPE FACTOR RELATIONS

SNO.	GEOMETRY	SHAPE FACTOR
4.	Finite, coaxial cylinders [Fig. E.4(a) and E.4(b)] $X = \frac{r_2}{r_1}, Y = \frac{L}{r_1}$ $A = X^2 + Y^2 - 1$ $B = Y^2 - X^2 + 1$	$F_{2-1} = \left(\frac{1}{X}\right) - \left(\frac{1}{\pi X}\right) \left\{ \cos^{-1} \left(\frac{B}{A} \right) - \left(\frac{1}{2Y} \right) \left[(A^2 + 4A - 4X^2 + 4)^{1/2} \cos^{-1} \left(\frac{B}{XA} \right) + B \sin^{-1} \left(\frac{1}{X} \right) - \frac{\pi A}{2} \right] \right\}$ $F_{2-2} = 1 - \left(\frac{1}{A} \right) \left(\frac{2}{\pi X} \right) \tan^{-1} \left[\frac{2(X^2 - 1)^{1/2}}{Y} \right] - \left(\frac{Y}{2\pi X} \right) \left\{ \left[\frac{(4X^2 + Y^2)}{Y} \right] \sin^{-1} \left[\frac{4(X^2 - 1) + \left(\frac{Y}{X} \right)^2 (X^2 - 2)}{Y^2 + 4(X^2 - 1)} \right] - \sin^{-1} \left[\frac{(X^2 - 2)}{X^2} \right] + \left(\frac{\pi}{2} \right) \left[\frac{(4X^2 + Y^2)^{1/2}}{Y - 1} \right] \right\}$
5.	Parallel, coaxial disks (Fig. E.5) $R_1 = \frac{r_1}{L}, R_2 = \frac{r_2}{L}$ $X = 1 + \frac{(1 + R_2^2)}{R_1^2}$	$F_{1-2} = \frac{\left[X - \left\{ X^2 - 4 \left(\frac{R_2}{R_1} \right)^2 \right\}^{1/2} \right]}{2}$

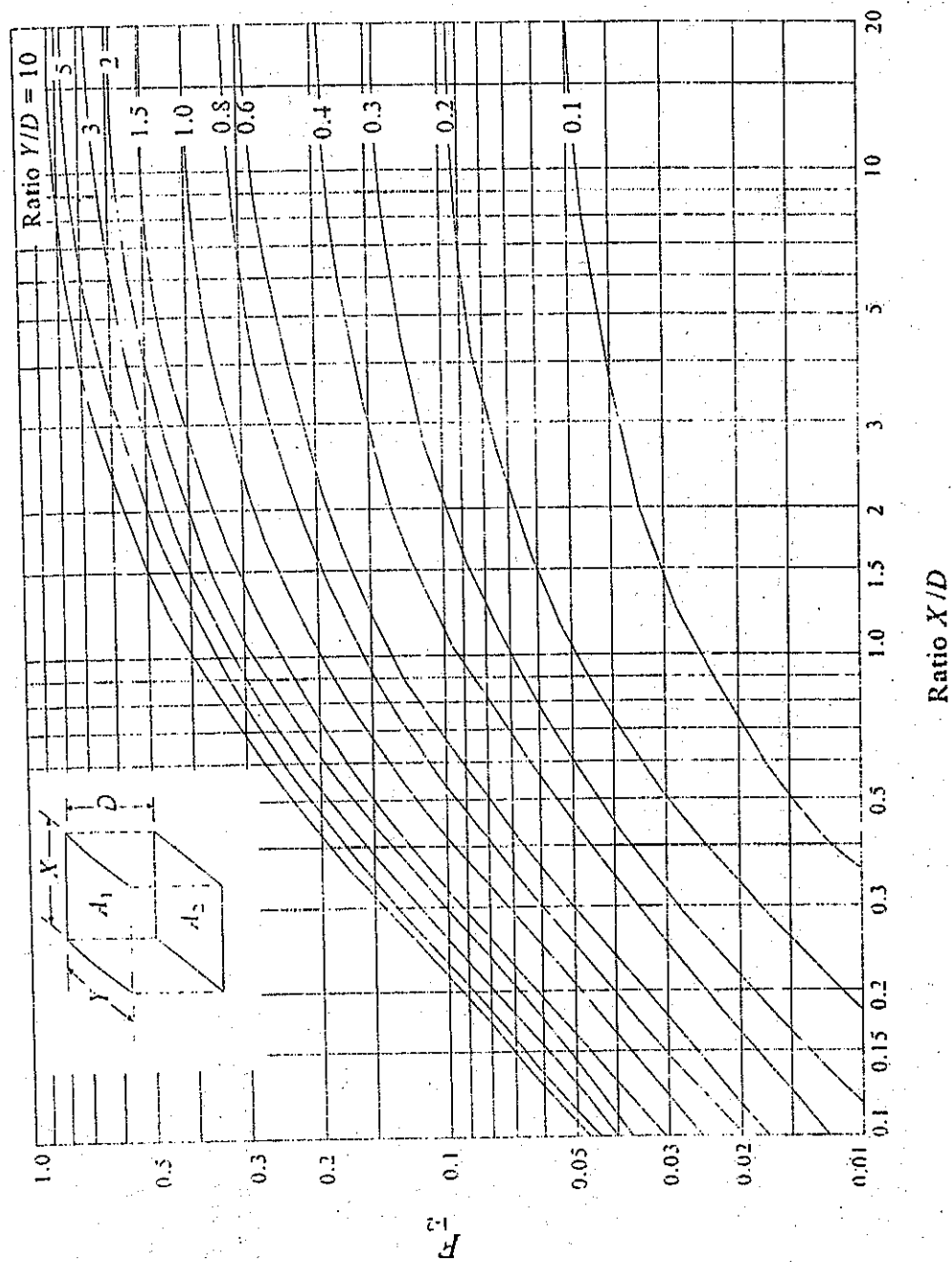


Fig. E.1 Radiation shape factor for radiation between parallel rectangles.

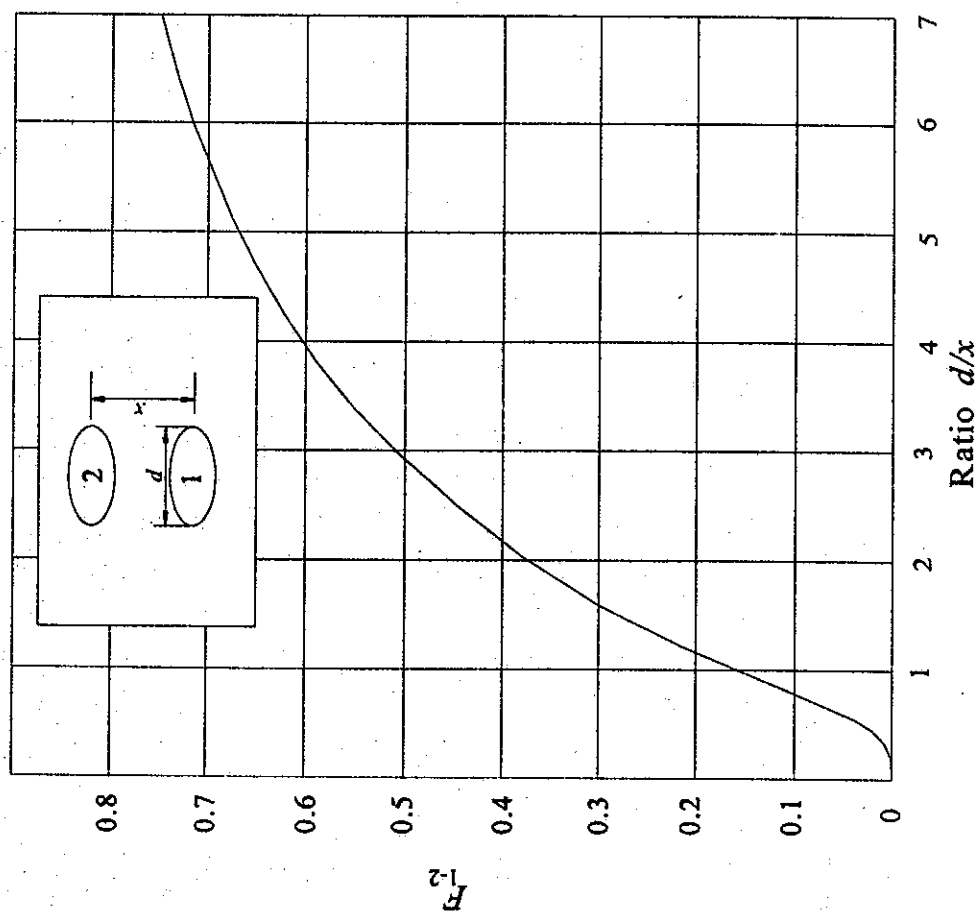


Fig. E.2 Radiation shape factor for radiation between parallel disks.

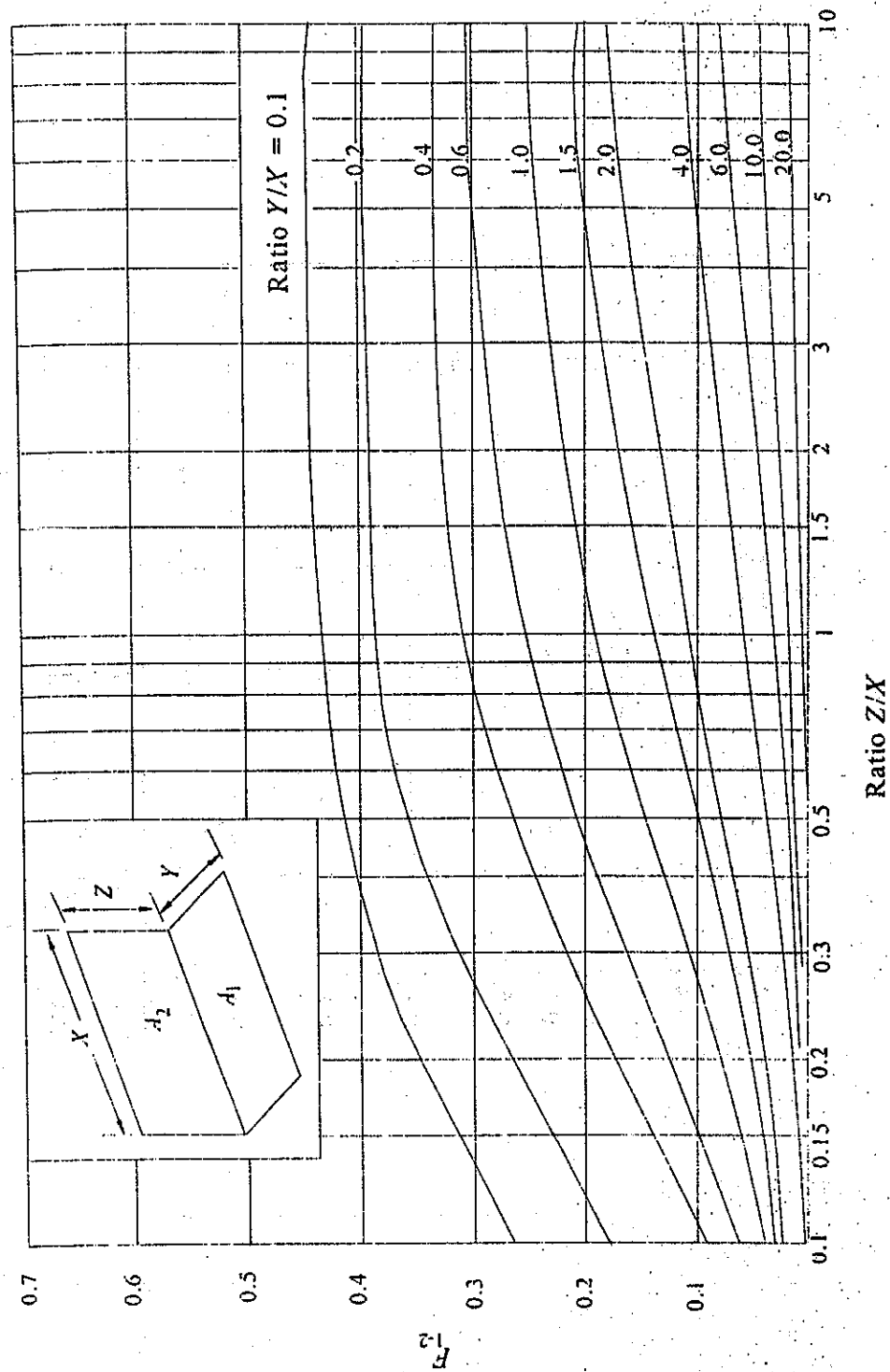


Fig. E.3 Radiation shape factor for radiation between perpendicular rectangles with a common edge.

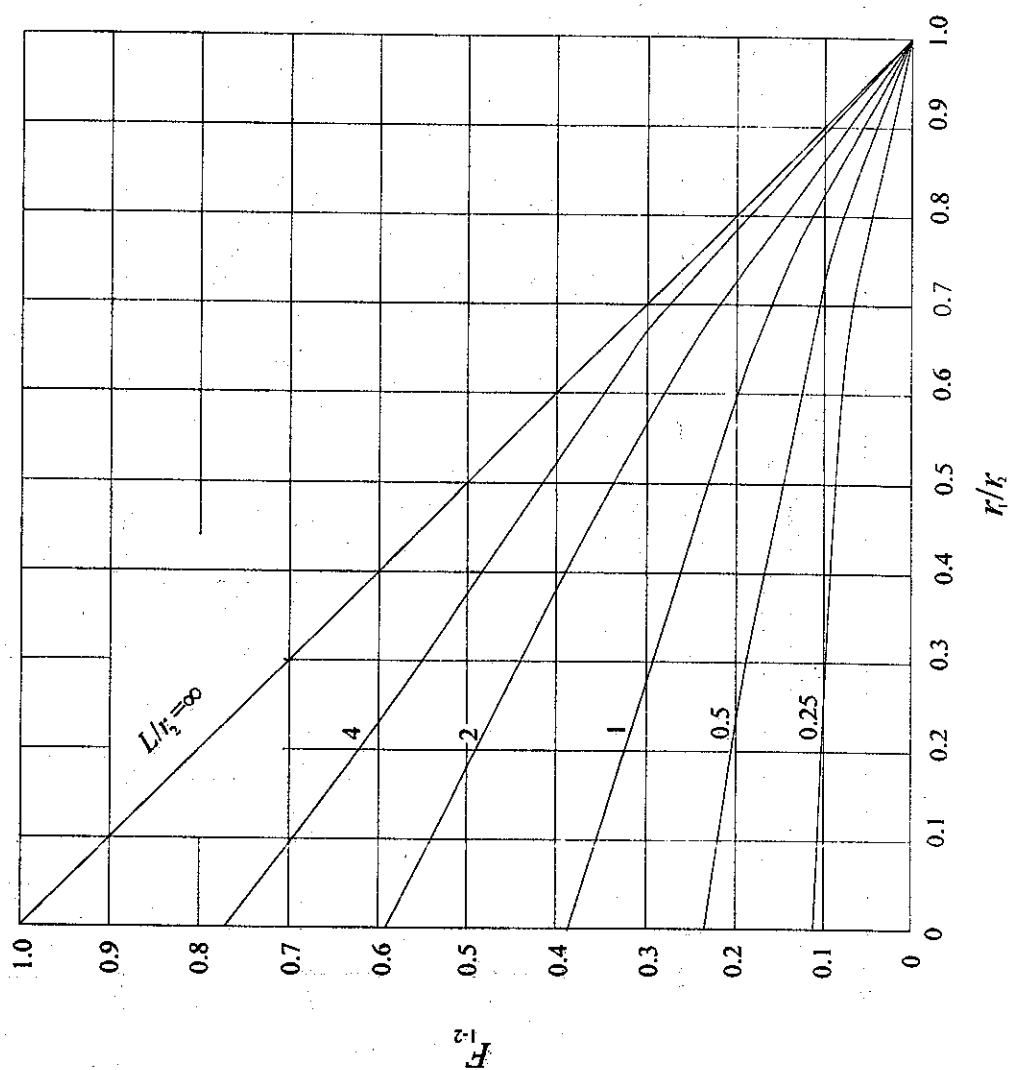


Fig. E.4 (a) Radiation shape factor for two concentric cylinders of finite length. Outer cylinder to itself.

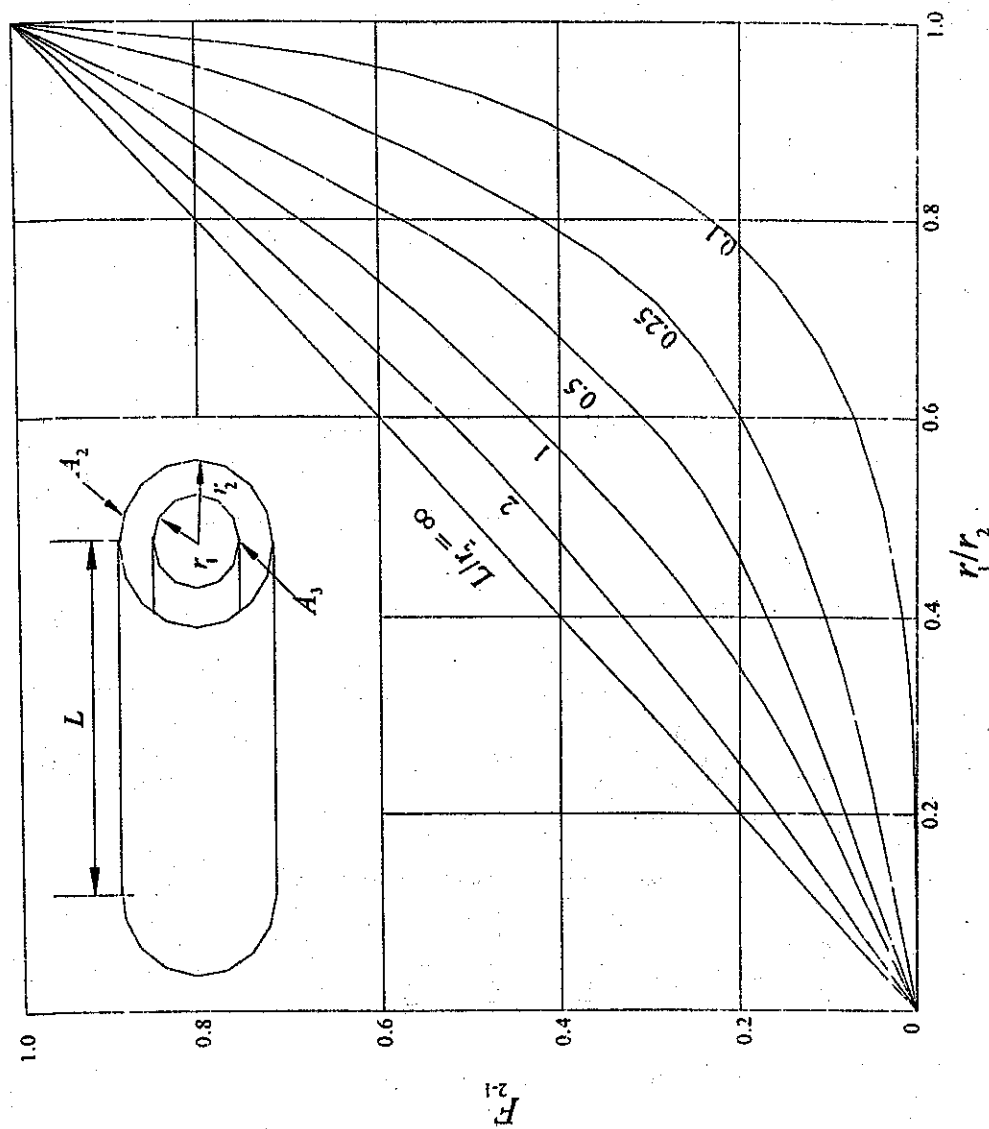


Fig. E.4 (b) Radiation shape factor for two concentric cylinders of finite length. Outer cylinder to inner cylinder.

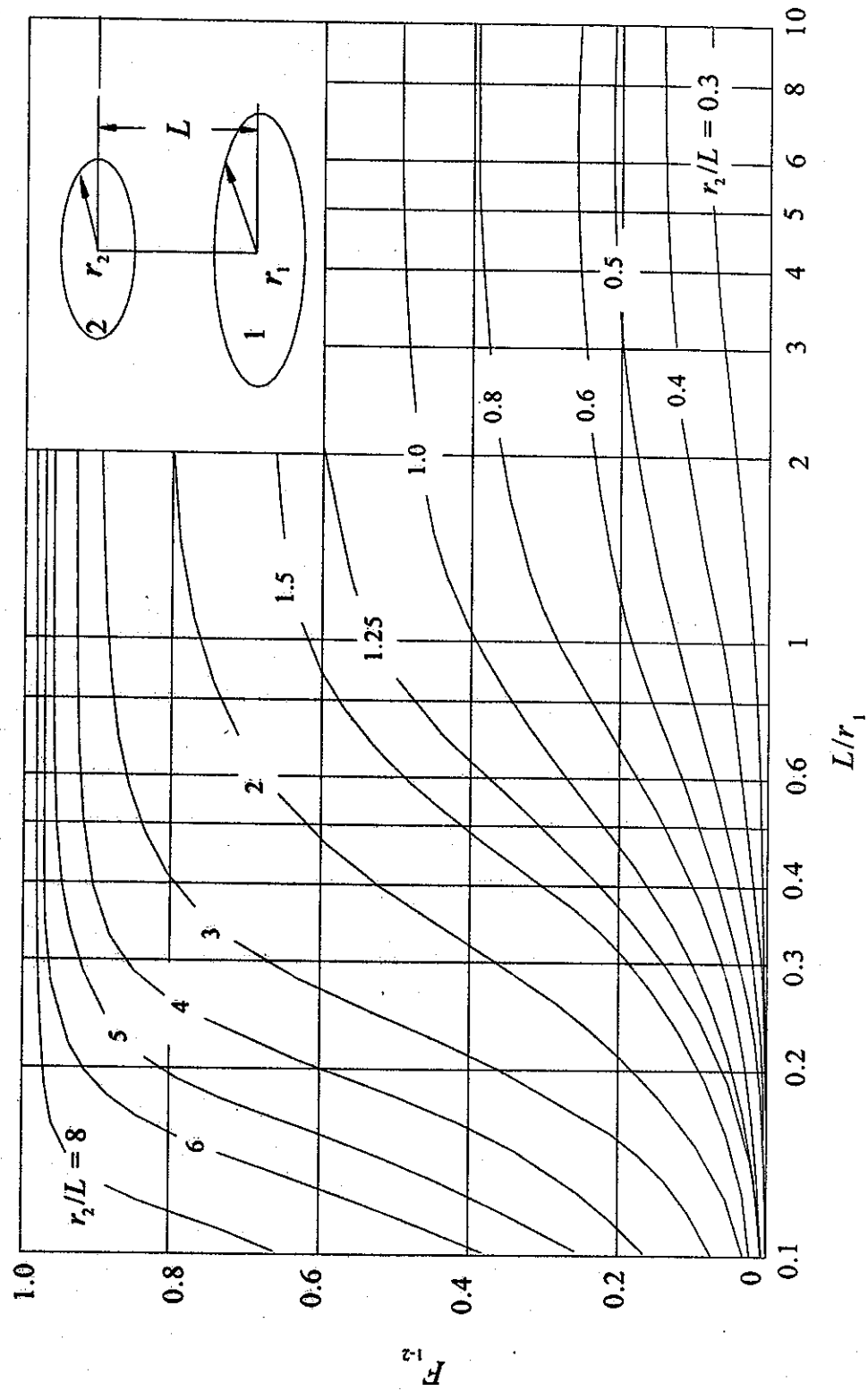


Fig. E.5 Radiation shape factor for radiation between two parallel concentric disks

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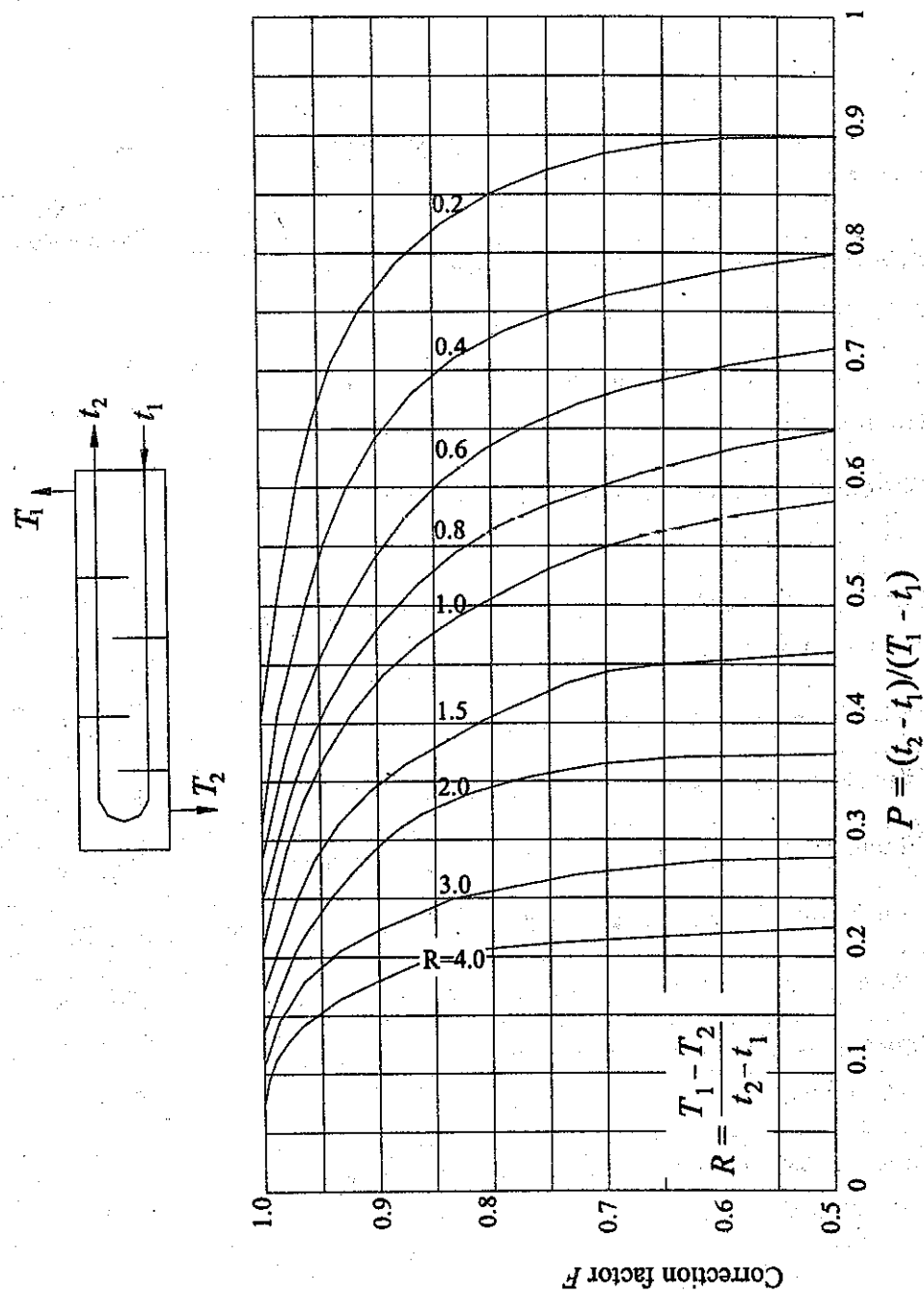


Fig. F.1 Correction factor plot for exchanger with one shell pass and two, four or any multiple of tube passes

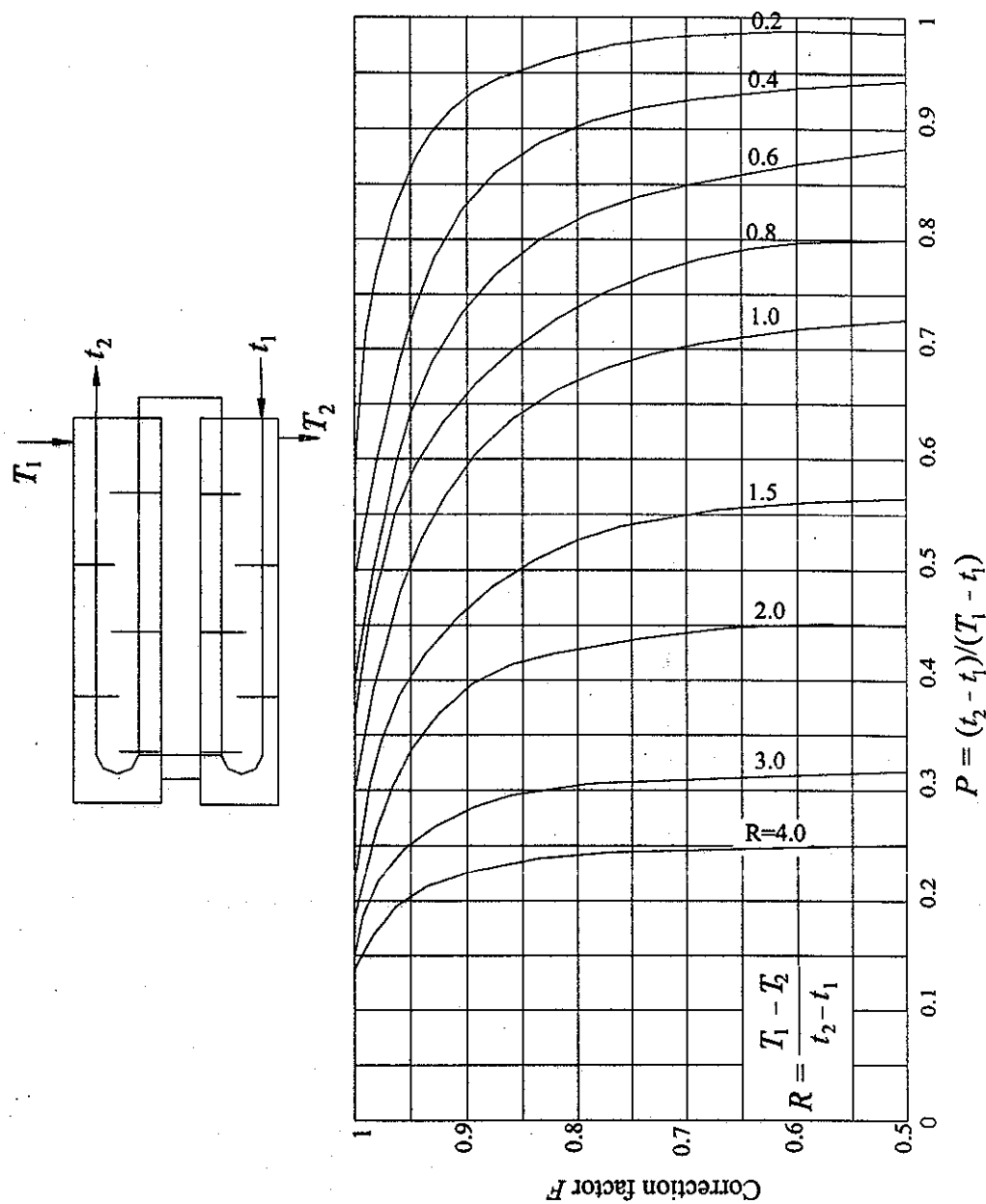


Fig F.2 Correction-factor plot for exchanger with two shell passes and four, eight or any multiple of tube passes.

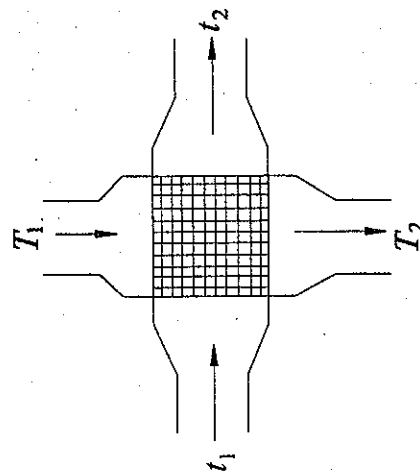
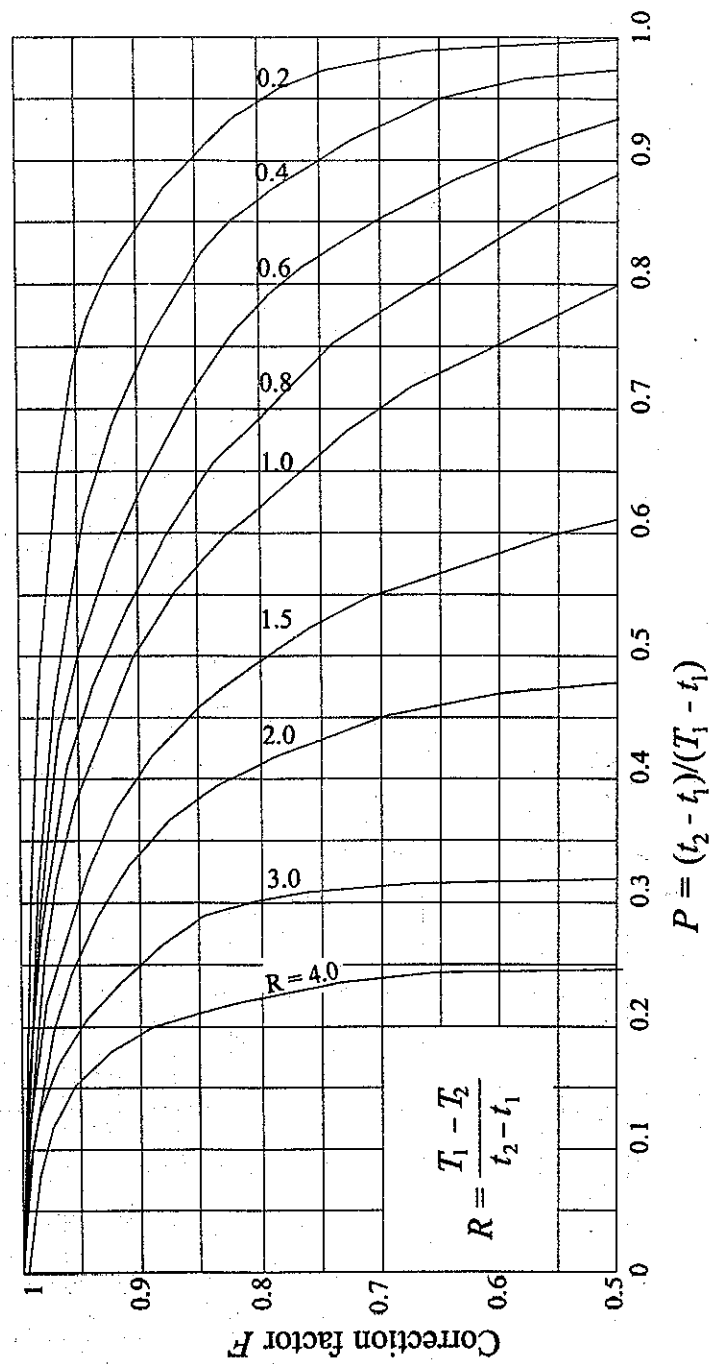


Fig. F.3 Correction-factor plot for single-pass cross flow exchanger, both fluids unmixed.

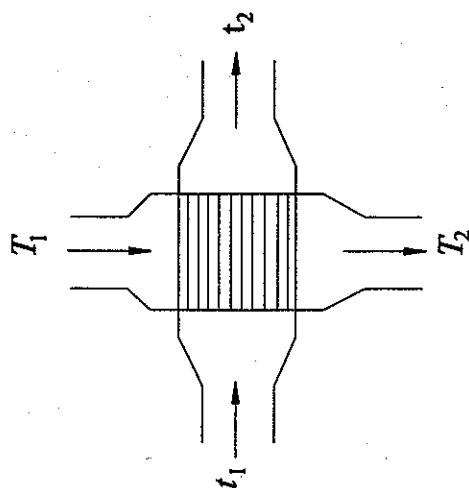
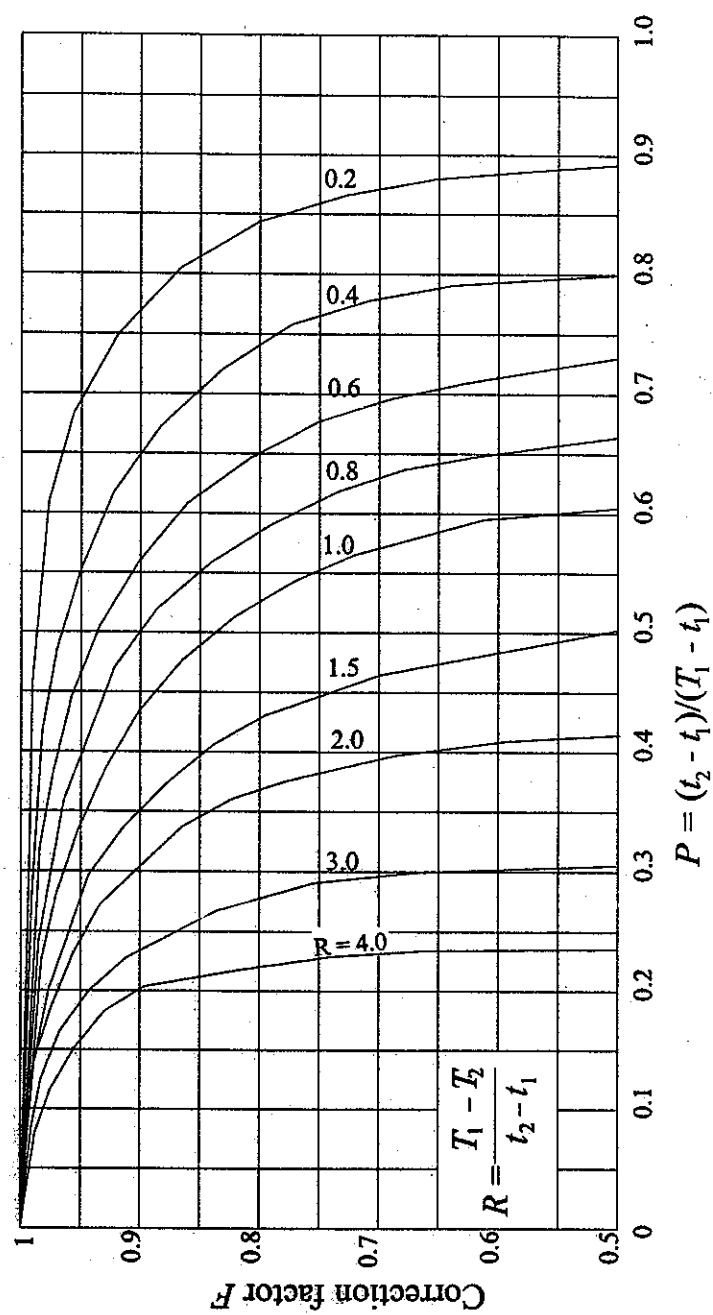


Fig. F.4 Correction-factor plot for single-pass cross flow exchanger, one fluid mixed, the other unmixed.

TABLE F.1
HEAT EXCHANGER EFFECTIVENESS RELATIONS

$$N = NTU = \frac{UA}{C_{\min}} \quad C = \frac{C_{\min}}{C_{\max}}$$

FLOW GEOMETRY		RELATION
Double pipe	Parallel flow	$\varepsilon = \frac{1 - \exp[-N(1+C)]}{1+C}$
	Counter flow	$\varepsilon = \frac{1 - \exp[-N(1-C)]}{1 - C \exp[-N(1-C)]}$
	Counter flow, $C=1$	$\varepsilon = \frac{N}{N+1}$
Cross Flow	Both fluids unmixed	$\varepsilon = 1 - \exp\left[\frac{\exp(-NCn) - 1}{Cn}\right]$ where $n = N^{-0.22}$
	Both fluids mixed	$\varepsilon = \left[\frac{1}{1 - \exp(-N)} + \frac{C}{1 - \exp(-NC)} - \frac{1}{N}\right]^{-1}$
	$C_{\max}$ mixed, $C_{\min}$ unmixed	$\varepsilon = (1/C)[1 - \exp[-C(1 - e^{-N})]]$
	$C_{\max}$ unmixed, $C_{\min}$ mixed	$\varepsilon = 1 - \exp[-(1/C)[1 - \exp(-NC)]]$
Shell and tube	One shell pass, 2, 4, 6, tube passes	$\varepsilon = 2 \left(1 + C + (1 + C^2)^{1/2} \times \frac{1 + \exp[-N(1 + C^2)^{1/2}]}{1 - \exp[-N(1 + C^2)^{1/2}]} \right)^{-1}$
	Multiple shell passes, 2n, 4n, 6n tube passes N = No. of shell passes	$\varepsilon = \frac{[(1 - \varepsilon_p C)/(1 - \varepsilon_p)]^n - 1}{[(1 - \varepsilon_p C)/(1 - \varepsilon_p)]^n - C}$

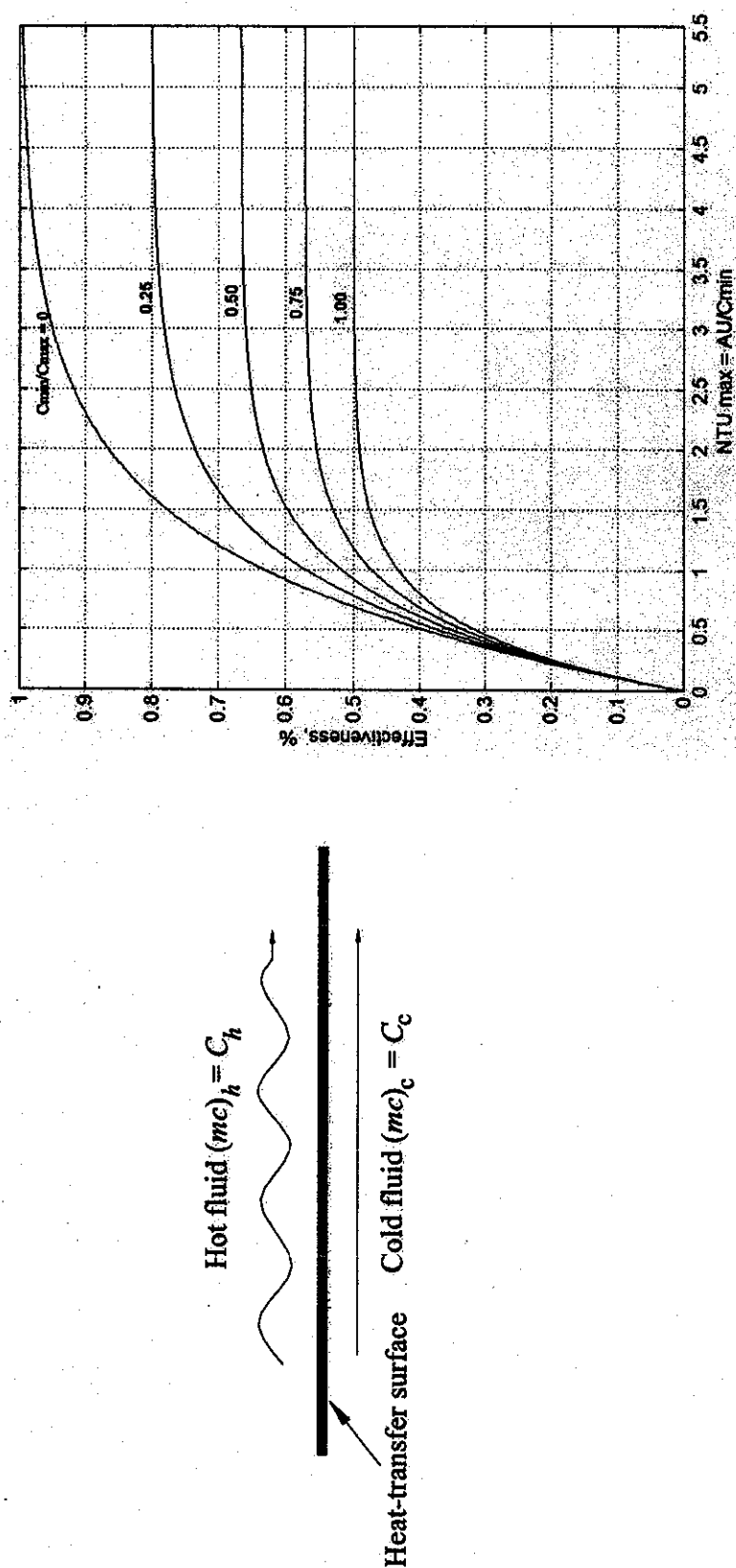


Fig. F.5 Effectiveness for parallel-flow exchanger performance.

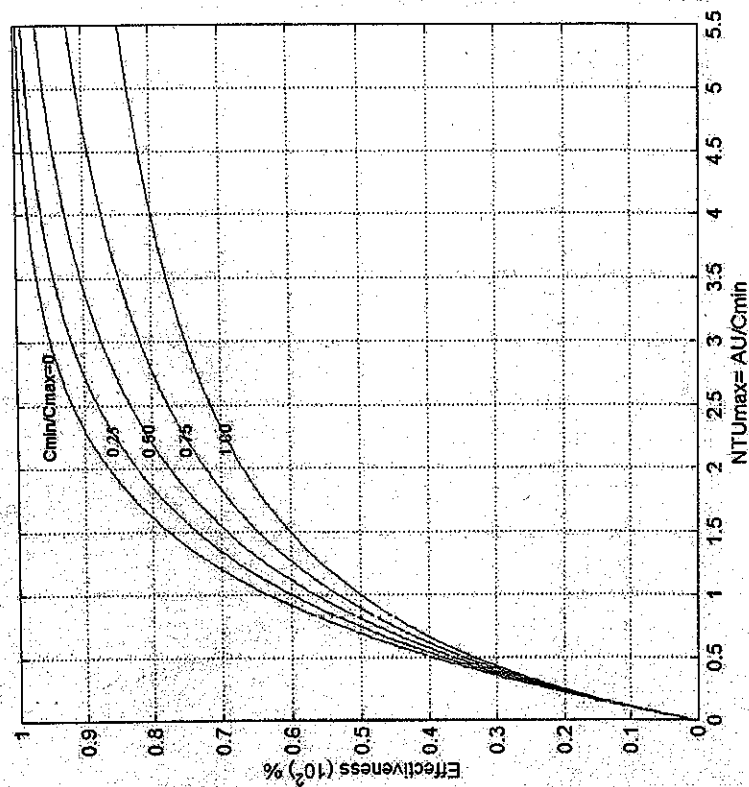
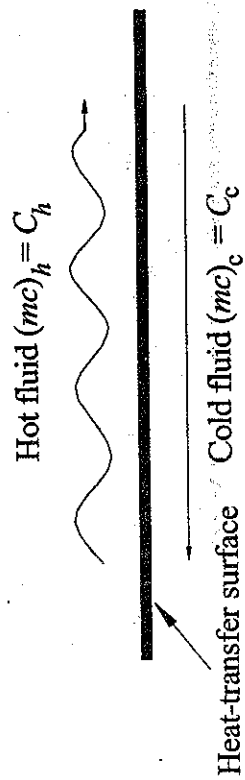


Fig. F.6 Effectiveness for counter flow heat exchanger performance.

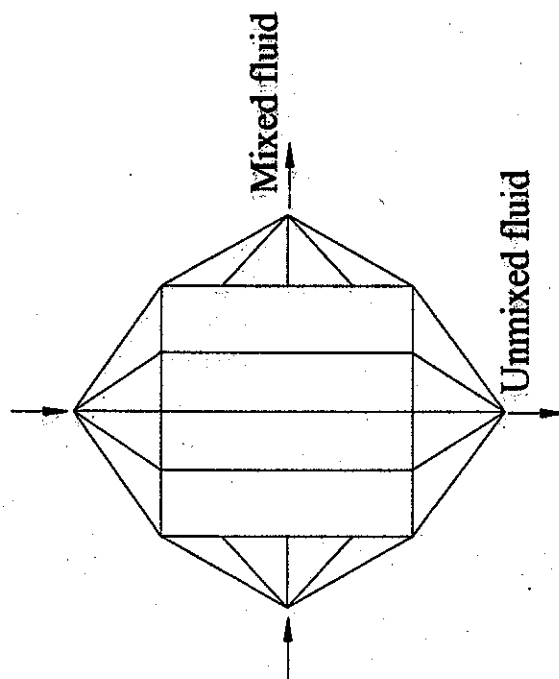
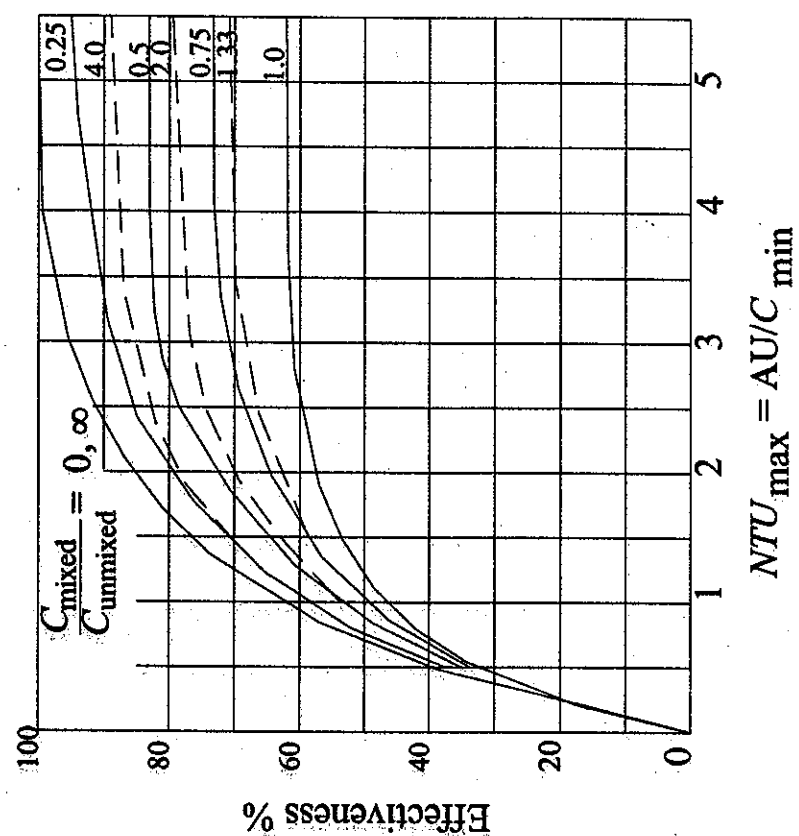


Fig. F.7 Effectiveness for cross-flow exchanger with one fluid mixed.

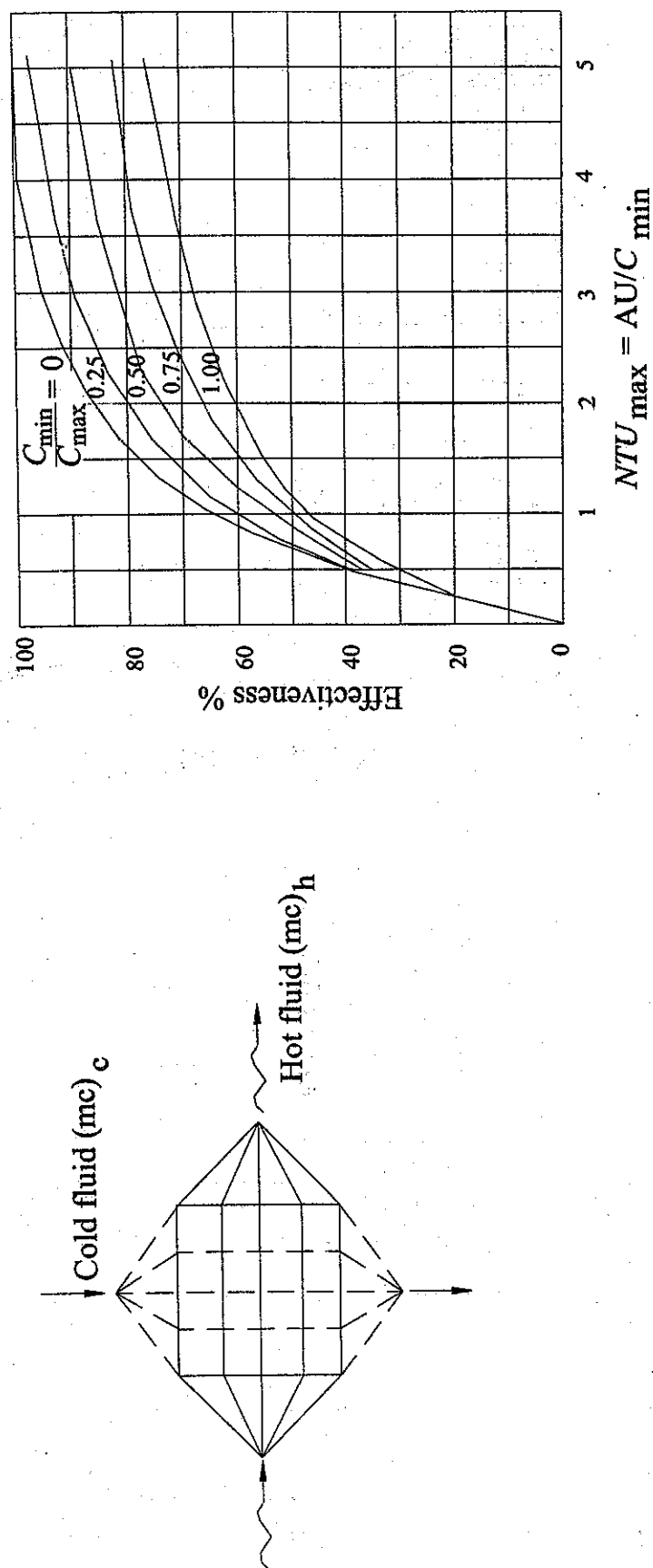


Fig. F.8 Effectiveness for cross-flow exchanger with fluids unmixed.

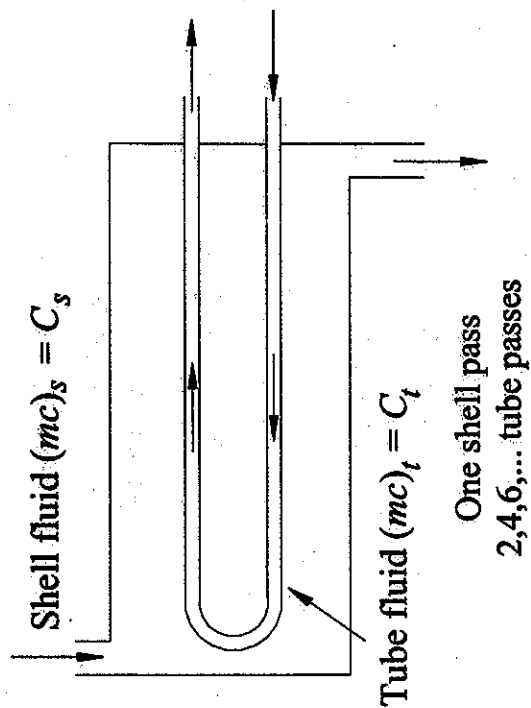
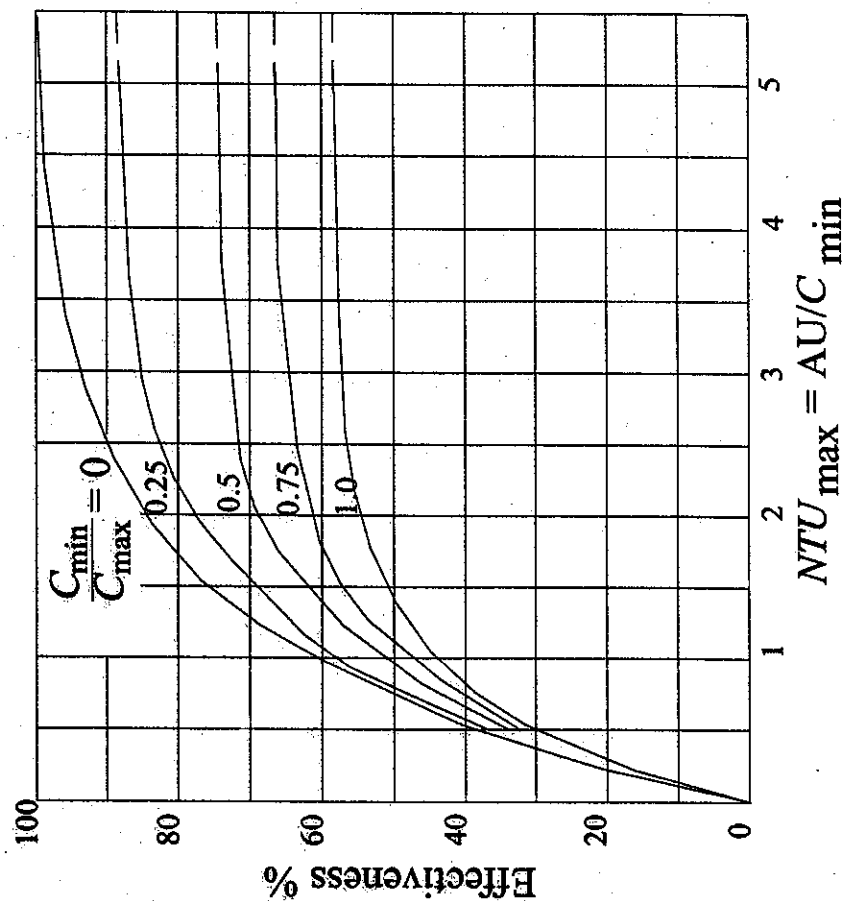


Fig. F.9 Effectiveness for 1-2 parallel counter flow exchanger performance.

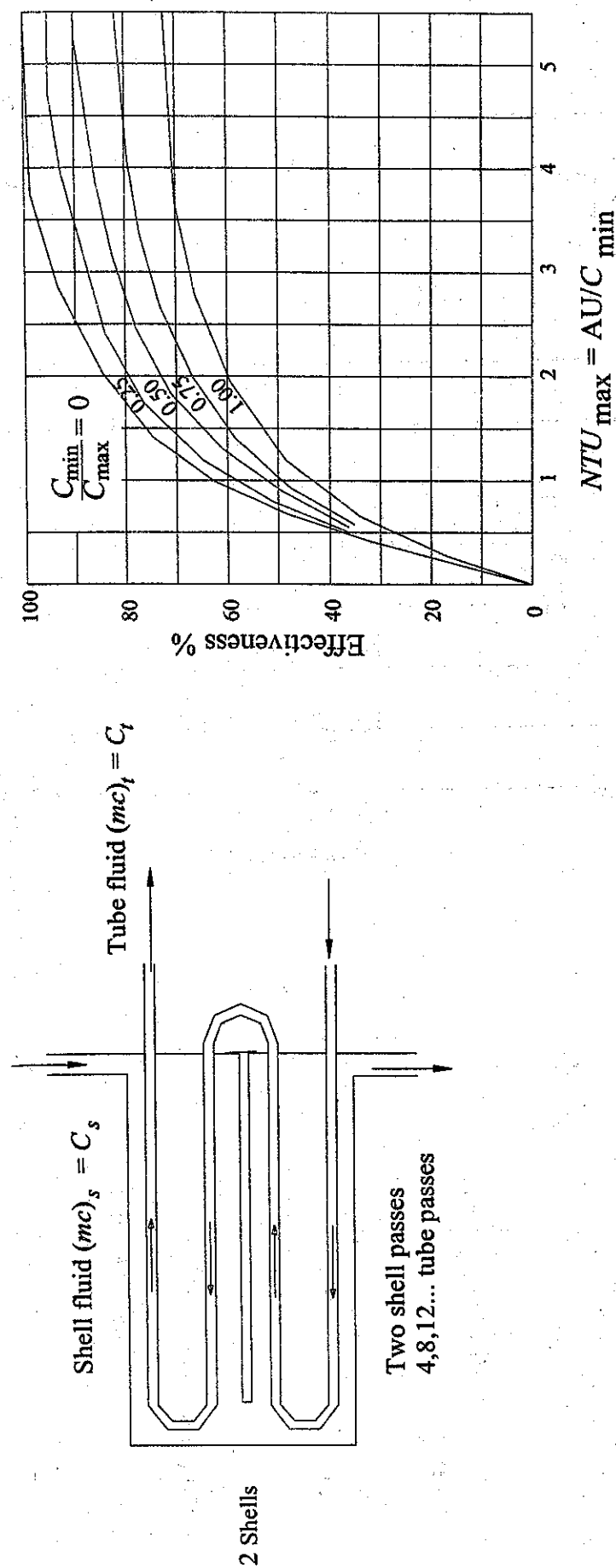


Fig. F.10 Effectiveness for 2-4 multi pass counter flow exchanger performance